

not intersect.²⁹ The nonexistence of competitive equilibrium can occur here for the same reason that we have already alluded to in small type in Section 10.C: The long-run production technologies we are considering exhibit nonconvexities.

It seems plausible, however, that when the efficient scale of a firm is small relative to the size of the market, this “integer problem” should not be too much of a concern. In fact, when we study oligopolistic markets in Chapter 12, we shall see that when firms’ efficient scales are small in this sense, the oligopolistic equilibrium price is close to $\bar{c}$, the equilibrium price we would derive if we simply ignored the integer constraint on the number of firms J^* . Intuitively, when the efficient scale is small, we will have many firms in the industry and the equilibrium, although not strictly competitive, will involve a price close to $\bar{c}$. Thus, if the efficient scale is small relative to the size of the market [as measured by $x(\bar{c})$], then ignoring the integer problem and treating firms as price takers gives approximately the correct answer.

Third, when an equilibrium exists, as in Figure 10.F.4, the equilibrium outcome maximizes Marshallian aggregate surplus and therefore is Pareto optimal. To see this, note from Figure 10.F.4 that aggregate surplus at the considered equilibrium is equal to

$$\text{Max}_{x \geq 0} \int_0^x P(s) ds - \bar{c}x,$$

the maximized value of aggregate surplus when firms’ cost functions are $\bar{c}q$. But because $c(q) \geq \bar{c}q$ for all q , this must be the largest attainable value of aggregate surplus given the actual cost function $c(\cdot)$; that is,

$$\text{Max}_{x \geq 0} \int_0^x P(s) ds - \bar{c}x \geq \int_0^{\hat{x}} P(s) ds - Jc(\hat{x}/J),$$

for all $\hat{x}$ and J . This fact provides an example of a point we raised at the end of Section 10.D (and will substantiate with considerable generality in Chapter 16): The first welfare theorem continues to be valid even in the absence of convexity of individual production sets.

Short-Run and Long-Run Comparative Statics

Although firms may enter and exit the market in response to profit opportunities in the long run, these changes may take time. For example, factories may need to be shut down, the workforce reduced, and machinery sold when a firm exits an industry. It may even pay a firm to continue operating until a suitable buyer for its plant and equipment can be found. When examining the comparative statics effects of a shock to a market, it is therefore important to distinguish between long-run and short-run effects.

Suppose, for example, that we are at a long-run equilibrium with J^* active firms

29. An intermediate case between constant returns (where any scale is efficient) and the case of a unique efficient scale occurs when there is a range $[\bar{q}, \tilde{q}]$ of efficient scales (the average cost curve has a flat bottom). In this case, the integer problem is mitigated. For a long-run competitive equilibrium to exist, we now only need there to be some $q \in [\bar{q}, \tilde{q}]$ such that $x(\bar{c})/q$ is an integer. Of course, as the interval $[\bar{q}, \tilde{q}]$ grows larger, not only are the chances of a long-run equilibrium existing greater, but so are the chances of indeterminacy of the equilibrium number of firms (i.e., of multiple equilibria involving differing numbers of firms).

each producing q^* units of output and that there is some shock to demand (similar points can be made for supply shocks). In the short run, it may be impossible for any new firms to organize and enter the industry, and so we will continue to have J^* firms for at least some period of time. Moreover, these J^* firms may face a short-run cost function $c_s(\cdot)$ that differs from the long-run cost function $c(\cdot)$ because various input levels may be fixed in the short run. For example, firms may have the long-run cost function

$$c(q) = \begin{cases} K + \psi(q) & \text{if } q > 0 \\ 0 & \text{if } q = 0, \end{cases} \quad (10.F.1)$$

where $\psi(0) = 0$, $\psi'(q) > 0$, and $\psi''(q) > 0$. But in the short run, it may be impossible for an active firm to recover its fixed costs if it exits and sets $q = 0$. Hence, in the short run the firm has the cost function

$$c_s(q) = K + \psi(q) \quad \text{for all } q \geq 0. \quad (10.F.2)$$

Another possibility is that $c(q)$ might be the cost function of some multiple-input production process, and in the short run an active firm may be unable to vary its level of some inputs. (See the discussion in Section 5.B on this point and also Exercises 10.F.5 and 10.F.6 for illustrations.)

Whenever the distinction between short run and long run is significant, the *short-run comparative statics effects* of a demand shock may best be determined by solving for the competitive equilibrium given J^* firms, each with cost function $c_s(\cdot)$, and the new demand function. This is just the equilibrium notion studied in Section 10.C, where we take firms' cost functions to be $c_s(\cdot)$. The *long-run comparative statics effects* can then be determined by solving for the long-run (i.e., free entry) equilibrium given the new demand function and long-run cost function $c(\cdot)$.

Example 10.F.1: Short-Run and Long-Run Comparative Statics with Lumpy Fixed Costs that Are Sunk in the Short Run. Suppose that the long-run cost function $c(\cdot)$ is given by (10.F.1) but that in the short run the fixed cost K is sunk so that $c_s(\cdot)$ is given by (10.F.2). The aggregate demand function is initially $x(\cdot, \alpha_0)$, and the industry is at a long-run equilibrium with J_0 firms, each producing $\bar{q}$ units of output [the efficient scale for cost function $c(\cdot)$], and a price of $p^* = \bar{c} = c(\bar{q})/\bar{q}$. This equilibrium position is depicted in Figure 10.F.5.

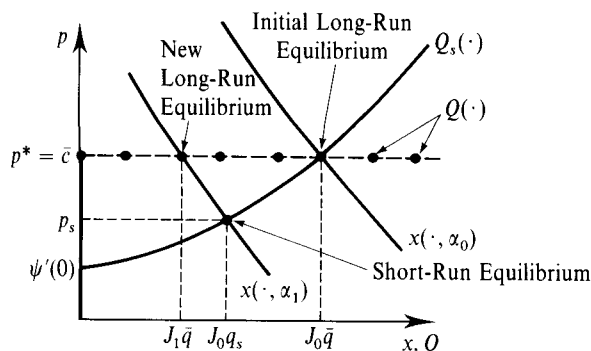


Figure 10.F.5
Short-run and long-run comparative statics in Example 10.F.1.

Now suppose that we have a shift to the demand function $x(\cdot, \alpha_1)$ shown in Figure 10.F.5. The short-run equilibrium is determined by the intersection of the graph of this demand function with the graph of the industry supply correspondence of the J_0 firms, each of which has short-run cost function $c_s(\cdot)$. The short-run aggregate supply correspondence is depicted as $Q_s(\cdot)$ in the figure. Thus, in the short run, the shock to demand causes price to fall to p_s and output per firm to fall to q_s . Firms' profits also fall; since $p_s < \bar{c}$, active firms lose money in the short run.

In the long run, however, firms exit in response to the decrease in demand, with the number of firms falling to $J_1 < J_0$, each producing output $\bar{q}$. Price returns to $p^* = \bar{c}$, aggregate consumption is $x(\bar{c}, \alpha_1)$, and all active firms once again earn zero profits. This new long-run equilibrium is also shown in Figure 10.F.5. ■

This division of dynamic adjustment into two periods, although useful as a first approximation, is admittedly crude. It may often be reasonable to think that there are several distinct short-run stages corresponding to different levels of adjustment costs associated with different decisions: in the very short run, production may be completely fixed; in the medium run, some inputs may be adjusted while others may not be; perhaps entry and exit take place only in the "very long run." Moreover, the methodology that we have discussed treats the two periods in isolation from each other. This approach ignores, for example, the possibility of intertemporal substitution by consumers when tomorrow's price is expected to differ from today's (intertemporal substitution might be particularly important for very short-run periods when the fact that many production decisions are fixed can make prices very sensitive to demand shocks).

These weaknesses are not flaws in the competitive model per se, but rather only in the somewhat extreme methodological simplification adopted here. A fully satisfactory treatment of these issues requires an explicitly dynamic model that places expectations at center stage. In Chapter 20 we study dynamic models of competitive markets in greater depth. Nevertheless, this simple dichotomization into long-run and short-run periods of adjustment is often a useful starting point for analysis.

10.G Concluding Remarks on Partial Equilibrium Analysis

In principle, the analysis of Pareto optimal outcomes and competitive equilibria requires the simultaneous consideration of the entire economy (a task we undertake in Part IV). Partial equilibrium analysis can be thought of as facilitating matters on two accounts. On the positive side, it allows us to determine the equilibrium outcome in the particular market under study in isolation from all other markets. On the normative side, it allows us to use Marshallian aggregate surplus as a welfare measure that, in many cases of interest, has a very convenient representation in terms of the area lying vertically between the aggregate demand and supply curves.

In the model considered in Sections 10.C to 10.F, the validity of both of these simplifications rested, implicitly, on two premises: first, that the prices of all commodities other than the one under consideration remain fixed; second, that there are no wealth effects in the market under study. We devote this section to a few additional interpretative comments regarding these assumptions. (See also Section 15.E for an example illustrating the limits of partial equilibrium analysis.)

The assumption that the prices of goods other than the good under consideration (say, good ℓ) remain fixed is essential for limiting our positive and normative analysis to a single market. In Section 10.B, we justified this assumption in terms of the market for good ℓ being small and having a diffuse influence over the remaining markets. However, this is not its only possible justification. For example, the nonsubstitution theorem (see Appendix A of Chapter 5) implies that the prices of all other goods will remain fixed if the numeraire is the only primary (i.e., nonproduced) factor, all produced goods other than ℓ are produced under conditions of constant returns using the numeraire and produced commodities other than ℓ as inputs, and there is no joint production.³⁰

Even when we cannot assume that all other prices are fixed, however, a generalization of our single-market partial equilibrium analysis is sometimes possible. Often we are interested not in a single market but in a group of commodities that are strongly interrelated either in consumers' tastes (tea and coffee are the classic examples) or in firms' technologies. In this case, studying one market at a time while keeping other prices fixed is no longer a useful approach because what matters is the *simultaneous* determination of *all* prices in the group. However, if the prices of goods outside the group may be regarded as unaffected by changes within the markets for this group of commodities, and if there are no wealth effects for commodities in the group, then we can extend much of the analysis presented in Sections 10.C to 10.F.

To this effect, suppose that the group is composed of M goods, and let $x_i \in \mathbb{R}_+^M$ and $q_j \in \mathbb{R}^M$ be vectors of consumptions and productions for these M goods. Each consumer has a utility function of the form

$$u_i(m_i, x_i) = m_i + \phi_i(x_i),$$

where m_i is the consumption of the numeraire commodity (i.e., the total expenditure on commodities outside the group). Firms' cost functions are $c_j(q_j)$. With this specification, many of the basic results of the previous sections go through unmodified (often it is just a matter of reinterpreting x_i and q_j as vectors). In particular, the results discussed in Section 10.C on the uniqueness of equilibrium and its independence from initial endowments still hold (see Exercise 10.G.1), as do the welfare theorems of Section 10.D. However, our ability to conduct welfare analysis using the areas lying vertically between demand and supply curves becomes much more limited. The cross-effects among markets with changing and interrelated prices cannot be

30. A simple example of this result arises when all produced goods other than ℓ are produced directly from the numeraire with constant returns to scale. In this case, the equilibrium price of each of these goods is equal to the amount of the numeraire that must be used as an input in its production per unit of output produced. More generally, prices for produced goods other than ℓ will remain fixed under the conditions of the nonsubstitution theorem because all efficient production vectors can be generated using a single set of techniques. In any equilibrium, the price of each produced good other than ℓ must be equal to the amount of the numeraire embodied in a unit of the good in the efficient production technique, either directly through the use of the numeraire as an input or indirectly through the use as inputs of produced goods other than ℓ that are in turn produced using the numeraire (or using other produced goods that are themselves produced using the numeraire, and so on).

ignored.³¹ (Exercises 10.G.3 to 10.G.5 ask you to consider some issues related to this point.)

The assumption of no wealth effects for good ℓ , on the other hand, is critical for the validity of the style of welfare analysis that we have carried out in this chapter. Without it, as we shall see in Part IV, Pareto optimality cannot be determined independently from the particular distribution of welfare sought, and we already know from Section 3.I that area measures calculated from Walrasian demand functions are not generally correct measures of compensating or equivalent variations (for which the Hicksian demand functions should be used). However, the assumption of no wealth effects is much less critical for positive analysis (determination of equilibrium, comparative statics effects, and so on). Even with wealth effects, the demand-and-supply apparatus can still be quite helpful for the positive part of the theory. The behavior of firms, for example, is not changed in any way. Consumers, on the other hand, have a demand function that, with prices of the other goods kept fixed, now depends only on the price for good ℓ and wealth. If wealth is determined from initial endowments and shareholdings, then we can view wealth as itself a function of the price of good ℓ (recall that other prices are fixed), and so we can again express demand as a function of this good's price alone. Formally, the analysis reduces to that presented in Section 10.C: The equilibrium in market ℓ can be identified as an intersection point of demand and supply curves.³²

31. A case in which the single-market analysis for good ℓ is still fully justified is when utility and cost functions have the form

$$u_i(m_i, x_i) = m_i + \phi_{\ell i}(x_{\ell i}) + \phi_{-\ell, i}(x_{-\ell, i}),$$

and

$$c_j(q_j) = c_{\ell j}(q_{\ell j}) + c_{-\ell, j}(q_{-\ell, j}),$$

where $x_{-\ell, i}$ and $q_{-\ell, j}$ are consumption and production vectors for goods in the group other than ℓ . With this additive separability in good ℓ , the markets for goods in the group other than ℓ do not influence the equilibrium price in market ℓ . Good ℓ is effectively independent of the group, and we can treat it in isolation, as we have done in the previous sections. (In point of fact, we do not even need to assume that the remaining markets in the group keep their prices fixed. What happens in them is simply irrelevant for equilibrium and welfare analysis in the market for good ℓ .) See Exercise 10.G.2.

32. The presence of wealth effects can lead, however, to some interesting new phenomena on the consumer's side. One is the *backward-bending* demand curve, where demand for a good is *increasing* in its price over some range. This can happen if consumers have endowments of good ℓ , because then an increase in its price increases consumers' wealth and could lead to a net increase in their demands for good ℓ , even if it is a normal good.

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EXERCISES

10.B.1^B The concept defined in Definition 10.B.2 is sometimes known as *strong Pareto efficiency*. An outcome is *weakly Pareto efficient* if there is no alternative feasible allocation that makes *all* individuals *strictly* better off.

(a) Argue that if an outcome is strongly Pareto efficient, then it is weakly Pareto efficient as well.

(b) Show that if all consumers' preferences are continuous and strongly monotone, then these two notions of Pareto efficiency are equivalent for any *interior* outcome (i.e., an outcome in which each consumer's consumption lies in the interior of his consumption set). Assume for simplicity that $X_i = \mathbb{R}_+^L$ for all i .

(c) Construct an example where the two notions are not equivalent. Why is the strong monotonicity assumption important in (b)? What about interiority?

10.B.2^A Show that if allocation $(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*)$ and price vector $p^* \gg 0$ constitute a competitive equilibrium, then allocation $(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*)$ and price vector αp^* also constitute a competitive equilibrium for any scalar $\alpha > 0$.

10.C.1^B Suppose that consumer i 's preferences can be represented by the utility function $u_i(x_{1i}, \dots, x_{Li}) = \sum_{\ell} \log(x_{\ell i})$ (these are Cobb–Douglas preferences).

(a) Derive his demand for good ℓ . What is the wealth effect?

(b) Now consider a sequence of situations in which we proportionately increase both the number of goods and the consumer's wealth. What happens to the wealth effect in the limit?

10.C.2^B Consider the two-good quasilinear model presented in Section 10.C with one consumer and one firm (so that $I = 1$ and $J = 1$). The initial endowment of the numeraire is $\omega_m > 0$, and the initial endowment of good ℓ is 0. Let the consumer's quasilinear utility function be $\phi(x) + m$, where $\phi(x) = \alpha + \beta \ln x$ for some $(\alpha, \beta) \gg 0$. Also, let the firm's cost function be $c(q) = \sigma q$ for some scalar $\sigma > 0$. Assume that the consumer receives all the profits of the firm. Both the firm and the consumer act as price takers. Normalize the price of good m to equal 1, and denote the price of good ℓ by p .

(a) Derive the consumer's and the firm's first-order conditions.

(b) Derive the competitive equilibrium price and output of good ℓ . How do these vary with α , β , and σ ?

10.C.3^B Consider a central authority who operates J firms with differentiable convex cost functions $c_j(q_j)$ for producing good ℓ from the numeraire. Define $C(q)$ to be the central authority's minimized cost level for producing aggregate quantity q ; that is

$$C(q) = \underset{(q_1, \dots, q_J) \geq 0}{\text{Min}} \sum_{j=1}^J c_j(q_j) \\ \text{s.t. } \sum_{j=1}^J q_j \geq q.$$

(a) Derive the first-order conditions for this cost-minimization problem.

(b) Show that at the cost-minimizing production allocation $(q_1^*, \dots, q_J^*)$, $C'(q) = c'_j(q_j^*)$ for all j with $q_j^* > 0$ (i.e., the central authority's marginal cost at aggregate output level q equals each firm's marginal cost level at the optimal production allocation for producing q).

(c) Show that if firms all maximize profit facing output price $p = C'(q)$ (with the price of the numeraire equal to 1), then the consequent output choices result in an aggregate output of q . Conclude that $C'(\cdot)$ is the inverse of the industry supply function $q(\cdot)$.

10.C.4^B Consider a central authority who has x units of good ℓ to allocate among I consumers, each of whom has a quasilinear utility function of the form $\phi_i(x_i) + m_i$, with $\phi_i(\cdot)$ a differentiable, increasing, and strictly concave function. The central authority allocates good ℓ to maximize the sum of consumers' utilities $\sum_i u_i$.

(a) Set up the central authority's problem and derive its first-order condition.

(b) Let $\gamma(x)$ be the value function of the central authority's problem, and let $P(x) = \gamma'(x)$ be its derivative. Show that if $(x_1^*, \dots, x_I^*)$ is the optimal allocation of good ℓ given available quantity x , then $P(x) = \phi_i'(x_i^*)$ for all i with $x_i^* > 0$.

(c) Argue that if all consumers maximize utility facing a price for good ℓ of $P(x)$ (with the price of the numeraire equal to 1), then the aggregate demand for good ℓ is exactly x . Conclude that $P(\cdot)$ is, in fact, the inverse of the aggregate demand function $x(\cdot)$.

10.C.5^B Derive the differential change in the equilibrium price in response to a differential change in the tax in Example 10.C.1 by applying the implicit function theorem to the system of equations (10.C.4) to (10.C.6).

10.C.6^B A tax is to be levied on a commodity bought and sold in a competitive market. Two possible forms of tax may be used: In one case, a *specific* tax is levied, where an amount t is paid per unit bought or sold (this is the case considered in the text); in the other case, an *ad valorem* tax is levied, where the government collects a tax equal to τ times the amount the seller receives from the buyer. Assume that a partial equilibrium approach is valid.

(a) Show that, with a specific tax, the ultimate cost of the good to consumers and the amounts purchased are independent of whether the consumers or the producers pay the tax.

(b) Show that this is not generally true with an ad valorem tax. In this case, which collection method leads to a higher cost to consumers? Are there special cases in which the collection method is irrelevant with an ad valorem tax?

10.C.7^B An ad valorem tax of τ (see Exercise 10.C.6 for a definition) is to be levied on consumers in a competitive market with aggregate demand curve $x(p) = Ap^\epsilon$, where $A > 0$ and $\epsilon < 0$, and aggregate supply curve $q(p) = \alpha p^\gamma$, where $\alpha > 0$ and $\gamma > 0$. Calculate the percentage change in consumer cost and producer receipts per unit sold for a small ("marginal") tax. Denote $\kappa = (1 + \tau)$. Assume that a partial equilibrium approach is valid.

Compute the elasticity of the equilibrium price with respect to κ . Argue that when $\gamma = 0$ producers bear the full effect of the tax while consumers' total costs of purchase are unaffected, and that when $\epsilon = 0$ it is consumers who bear the full burden of the tax. What happens when each of these elasticities approaches ∞ in absolute value?

10.C.8^B Suppose that there are J firms producing good ℓ , each with a differentiable cost function $c(q, \alpha)$ that is strictly convex in q , where α is an exogenous parameter that affects costs (it could be a technological parameter or an input price). Assume that $\partial c(q, \alpha) / \partial \alpha > 0$. The differentiable aggregate demand function for good ℓ is $x(p)$, with $x'(\cdot) \leq 0$. Assume that partial equilibrium analysis is justified.

Let $q^*(\alpha)$ be the *per firm* output and $p^*(\alpha)$ be the equilibrium price in the competitive equilibrium given α .

(a) Derive the marginal change in a firm's profits with respect to α .

(b) Give the weakest possible sufficient condition, stated in terms of marginal and average costs and/or their derivatives, that guarantees that if α increases marginally, then firms' equilibrium profits decline for any demand function $x(\cdot)$ having $x'(\cdot) \leq 0$. Show that if this condition is not satisfied, then there are demand functions such that profits increase when α increases.

(c) In the case where α is the price of factor input k , interpret the condition in (b) in terms of the conditional factor demand for input k .

10.C.9^B Suppose that in a partial equilibrium context there are J identical firms that produce good ℓ with cost function $c(w, q)$, where w is a vector of factor input prices. Show that an increase in the price of factor k , w_k , lowers the equilibrium price of good ℓ if and only if factor k is an *inferior* factor, that is, if at fixed input prices, the use of factor k is decreasing in a firm's output level.

10.C.10^B Consider a market with demand curve $x(p) = \alpha p^\varepsilon$ and with J firms, each of which has marginal cost function $c'(q) = \beta q^\eta$, where $(\alpha, \beta, \eta) \gg 0$ and $\varepsilon < 0$. Calculate the competitive equilibrium price and output levels. Examine the comparative statics change in these variables as a result of changes in α and β . How are these changes affected by ε and η ?

10.C.11^B Assume that partial equilibrium analysis is valid. Suppose that firms 1 and 2 are producing a positive level of output in a competitive equilibrium. The cost function for firm j is given by $c(q, \alpha_j)$, where α_j is an exogenous technological parameter. If α_1 differs from α_2 marginally, what is the difference in the two firms' profits?

10.D.1^B Prove that under the assumptions that the $\phi_i(\cdot)$ functions are strictly concave and the cost functions $c_j(\cdot)$ are convex, the optimal individual consumption levels of good ℓ in problem (10.D.2) are uniquely defined. Conclude that the optimal aggregate production level of good ℓ is therefore also uniquely defined. Show that if the cost functions $c_j(\cdot)$ are *strictly* convex, then the optimal individual production levels of good ℓ in problem (10.D.2) are also uniquely defined.

10.D.2^B Determine the optimal consumption and production levels of good ℓ for the economy described in Exercise 10.C.2. Compare these with the equilibrium levels you identified in that exercise.

10.D.3^B In the context of the two-good quasilinear economy studied in Section 10.D, show that any allocation that is a solution to problem (10.D.6) is Pareto optimal and that any Pareto optimal allocation is a solution to problem (10.D.6) for *some* choice of utility levels $(\bar{u}_2, \dots, \bar{u}_I)$.

10.D.4^B Derive the first-order conditions for problem (10.D.6) and compare them with conditions (10.D.3) to (10.D.5).

10.E.1^C Suppose that $J_d > 0$ of the firms that produce good ℓ are domestic firms, and $J_f > 0$ are foreign firms. All domestic firms have the same convex cost function for producing good ℓ , $c_d(q_j)$. All foreign firms have the same convex cost function $c_f(q_j)$. Assume that partial equilibrium analysis is valid.

The government of the domestic country is considering imposing a per-unit tariff of τ on imports of good ℓ . The government wants to maximize domestic welfare as measured by the *domestic* Marshallian surplus (i.e., the sum of domestic consumers' utilities less domestic firms' costs).

(a) Show that if $c_f(\cdot)$ is strictly convex, then imposition of a small tariff raises domestic welfare.

(b) Show that if $c_f(\cdot)$ exhibits constant returns to scale, then imposition of a small tariff lowers domestic welfare.

10.E.2^B Consumer surplus when consumers face effective price $\hat{p}$ can be written as

$$CS(\hat{p}) = \int_0^{x(\hat{p})} [P(s) - \hat{p}] ds.$$

Prove by means of a change of variables and integration by parts that this integral is equal to $\int_p^\infty x(s) ds$.

10.E.3^C (*Ramsey tax problem*) Consider a fully separable quasilinear model with L goods in which each consumer has preferences of the form $u_i(x_i) = x_{1i} + \sum_{\ell=2}^L \phi_{\ell i}(x_{\ell i})$ and each good $2, \dots, L$ is produced with constant returns to scale from good 1, using c_ℓ units of good 1 per unit of good ℓ produced. Assume that consumers initially hold endowments only of the numeraire, good 1. Hence, consumers are net sellers of good 1 to the firms and net purchasers of goods $2, \dots, L$.

In this setting, consumer i 's demand for each good $\ell \neq 1$ can be written in the form $x_{\ell i}(p_\ell)$, so that demand for good ℓ is independent of the consumer's wealth and all other prices, and welfare can be measured by the sum of the Marshallian aggregate surpluses in the $L - 1$ markets for nonnumeraire commodities (see Section 10.G and Exercise 10.G.2 for more on this).

Suppose that the government must raise R units of good 1 through (specific) commodity taxes. Note, in particular, that such taxes involve taxing a *transaction* of a good, *not* an individual's consumption level of that good.

Let t_ℓ denote the tax to be paid by a consumer in units of good 1 for each unit of good $\ell \neq 1$ purchased, and let t_1 be the tax in units of good 1 to be paid by consumers for each unit of good 1 sold to a firm. Normalize the price paid by firms for good 1 to equal 1. Under our assumptions, each choice of $t = (t_1, \dots, t_L)$ results in a consumer paying a total of $c_\ell + t_\ell$ per unit of good $\ell \neq 1$ purchased and having to part with $(1 + t_1)$ units of good 1 for each unit of good 1 sold to a firm.

(a) Consider two possible tax vectors t and t' . Show that if t' is such that $(c_\ell + t'_\ell) = \alpha(c_\ell + t_\ell)$ and $(1 + t'_1) = (1/\alpha)(1 + t_1)$ for some scalar $\alpha > 0$, then the two sets of taxes raise the same revenue. Conclude from this fact that the government can restrict attention to tax vectors that leave one good untaxed.

(b) Let good 1 be the untaxed good (i.e., set $t_1 = 0$). Derive conditions describing the taxes that should be set on goods $2, \dots, L$ if the government wishes to minimize the welfare loss arising from this taxation. Express this formula in terms of the elasticity of demand for each good.

(c) Under what circumstances should the tax rate on all goods be equal? In general, which goods should have higher tax rates? When would taxing only good 1 be optimal?

10.F.1^A Show that if $c(q)$ is strictly convex in q and $c(0) = 0$, then $\pi(p) > 0$ if and only if $p > c'(0)$.

10.F.2^B Consider a market with demand function $x(p) = A - Bp$ in which every potential firm has cost function $c(q) = K + \alpha q + \beta q^2$, where $\alpha > 0$ and $\beta > 0$.

(a) Calculate the long-run competitive equilibrium price, output per firm, aggregate output, and number of firms. Ignore the integer constraint on the number of firms. How does each of these vary with A ?

(b) Now examine the short-run competitive equilibrium response to a change in A starting from the long-run equilibrium you identified in (a). How does the change in price depend on the level of A in the initial equilibrium? What happens as $A \rightarrow \infty$? What accounts for this effect of market size?

10.F.3^B (D. Pearce) Consider a partial equilibrium setting in which each (potential) firm has a long-run cost function $c(\cdot)$, where $c(q) = K + \phi(q)$ for $q > 0$ and $c(0) = 0$. Assume that $\phi'(q) > 0$ and $\phi''(q) < 0$, and denote the firm's efficient scale by $\bar{q}$. Suppose that there is initially a long-run equilibrium with J^* firms. The government considers imposing two different types

of taxes: The first is an ad valorem tax of τ (see Exercise 10.C.6) on sales of the good. The second is a tax T that must be paid by any operating firm (where a firm is considered to be “operating” if it sells a positive amount). If the two taxes would raise an equal amount of revenue with the initial level of sales and number of firms, which will raise more after the industry adjusts to a new long-run equilibrium? (You should ignore the integer constraint on the number of firms.)

10.F.4^B (J. Panzar) Assume that partial equilibrium analysis is valid. The single-output, many-input technology for producing good ℓ has a differentiable cost function $c(w, q)$, where $w = (w_1, \dots, w_K)$ is a vector of factor input prices and q is the firm’s output of good ℓ . Given factor prices w , let $\bar{q}(w)$ denote the firm’s efficient scale. Assume that $\bar{q}(w) > 0$ for all w . Also let $p^*(w)$ denote the long-run equilibrium price of good ℓ when factor prices are w . Show that the function $p^*(w)$ is nondecreasing, homogeneous of degree one, and concave. (You should ignore the integer constraint on the number of firms.)

10.F.5^C Suppose that there are J firms that can produce good ℓ from K factor inputs with differentiable cost function $c(w, q)$. Assume that this function is strictly convex in q . The differentiable aggregate demand function for good ℓ is $x(p, \alpha)$, where $\partial x(p, \alpha)/\partial p < 0$ and $\partial x(p, \alpha)/\partial \alpha > 0$ (α is an exogenous parameter affecting demand). However, although $c(w, q)$ is the cost function when all factors can be freely adjusted, factor k cannot be adjusted in the short run.

Suppose that we are initially at an equilibrium in which all inputs are optimally adjusted to the equilibrium level of output q^* and factor prices w so that, letting $z_k(w, q)$ denote a firm’s conditional factor demand for input k when all inputs can be adjusted, $z_k^* = z_k(w, q^*)$.

(a) Show that a firm’s equilibrium response to an increase in the price of good ℓ is larger in the long run than in the short run.

(b) Show that this implies that the long-run equilibrium response of p_ℓ to a marginal increase in α is smaller than the short-run response. Show that the reverse is true for the response of the equilibrium aggregate consumption of good ℓ (hold the number of firms equal to J in both the short run and long run).

10.F.6^B Suppose that the technology for producing a good uses capital (z_1) and labor (z_2) and takes the Cobb–Douglas form $f(z_1, z_2) = z_1^\alpha z_2^{1-\alpha}$, where $\alpha \in (0, 1)$. In the long run, both factors can be adjusted; but in the short run, the use of capital is fixed. The industry demand function takes the form $x(p) = a - bp$. The vector of input prices is (w_1, w_2) . Find the long-run equilibrium price and aggregate quantity. Holding the number of firms and the level of capital fixed at their long-run equilibrium levels, what is the short-run industry supply function?

10.F.7^B Consider a case where in the short run active firms can increase their use of a factor but cannot decrease it. Show that the short-run cost curve will exhibit a kink (i.e., be nondifferentiable) at the current (long-run) equilibrium. Analyze the implications of this fact for the relative variability of short-run prices and quantities.

10.G.1^B Consider the case of an interrelated group of M commodities. Let consumer i ’s utility function take the form $u_i(x_{1i}, \dots, x_{Mi}) = m_i + \phi_i(x_{1i}, \dots, x_{Mi})$. Assume that $\phi_i(\cdot)$ is differentiable and strictly concave. Let firm j ’s cost function be the differentiable convex function $c_j(q_{1j}, \dots, q_{Mj})$.

Normalize the price of the numeraire to be 1. Derive $(I + J + 1)M$ equations characterizing the $(I + J + 1)M$ equilibrium quantities $(x_{1i}^*, \dots, x_{Mi}^*)$ for $i = 1, \dots, I$, $(q_{1j}^*, \dots, q_{Mj}^*)$ for $j = 1, \dots, J$, and $(p_1^*, \dots, p_M^*)$. [Hint: Derive consumers’ and firms’ first-order conditions and the $M - 1$ market-clearing conditions in parallel to our analysis of the single-market case.] Argue that the equilibrium prices and quantities of these M goods are independent of

consumers' wealths, that equilibrium individual consumptions and aggregate production levels are unique, and that if the $c_j(\cdot)$ functions are strictly convex, then equilibrium individual production levels are also unique.

10.G.2^B Consider the case in which the functions $\phi_i(\cdot)$ and $c_j(\cdot)$ in Exercise 10.G.1 are separable in good ℓ (one of the goods in the group): $\phi_i(\cdot) = \phi_{\ell i}(x_{\ell i}) + \phi_{-\ell, i}(x_{-\ell, i})$ and $c_j(\cdot) = c_{\ell j}(q_{\ell j}) + c_{-\ell, j}(q_{-\ell, j})$. Argue that in this case, the equilibrium price, consumption, and production of good ℓ can be determined independently of other goods in the group. Also argue that under the same assumptions as in the single-market case studied in Section 10.E, changes in welfare caused by changes in the market for this good can be captured by the Marshallian aggregate surplus for this good, $\sum_i \phi_{\ell i}(x_{\ell i}) - \sum_j c_{\ell j}(q_{\ell j})$, which can be represented in terms of the areas lying vertically between the demand and supply curves for good ℓ . Note the implication of these results for the case in which we have separability of all goods: $\phi_i(\cdot) = \sum_{\ell} \phi_{\ell i}(x_{\ell i})$ and $c_j(\cdot) = \sum_{\ell} c_{\ell j}(q_{\ell j})$.

10.G.3^B Consider a three-good economy ($\ell = 1, 2, 3$) in which every consumer has preferences that can be described by the utility function $u(x) = x_1 + \phi(x_2, x_3)$ and there is a single production process that produces goods 2 and 3 from good 1 having $c(q_2, q_3) = c_2 q_2 + c_3 q_3$. Suppose that we are considering a tax change in only a single market, say market 2.

(a) Show that if the price in market 3 is undistorted (i.e., if $t_3 = 0$), then the change in aggregate surplus caused by the tax change can be captured solely through the change in the area lying vertically between market 2's demand and supply curves holding the price of good 3 at its initial level.

(b) Show that if market 3 is initially distorted because $t_3 > 0$, then by using only the single-market measure in (a), we would overstate the decrease in aggregate surplus if good 3 is a substitute for good 2 and would understate it if good 3 is a complement. Provide an intuitive explanation of this result. What is the correct measure of welfare change?

10.G.4^B Consider a three-good economy ($\ell = 1, 2, 3$) in which every consumer has preferences that can be described by the utility function $u(x) = x_1 + \phi(x_2, x_3)$ and there is a single production process that produces goods 2 and 3 from good 1 having $c(q_2, q_3) = c_2 q_2 + c_3 q_3$. Derive an expression for the welfare loss from an increase in the tax rates on both goods.

10.G.5^B Consider a three-good economy ($\ell = 1, 2, 3$) in which every consumer has preferences that can be described by the utility function $u(x) = x_1 + \phi(x_2, x_3)$ and there is a single production process that produces goods 2 and 3 from good 1 having $c(q_2, q_3) = c_2(q_2) + c_3(q_3)$, where $c_2(\cdot)$ and $c_3(\cdot)$ are strictly increasing and strictly convex.

(a) If goods 2 and 3 are substitutes, what effect does an increase in the tax on good 2 have on the price paid by consumers for good 3? What if they are complements?

(b) What is the bias from applying the formula for welfare loss you derived in part (b) of Exercise 10.G.3 using the price paid by consumers for good 3 prior to the tax change in both the case of substitutes and that of complements?

Externalities and Public Goods

11.A Introduction

In Chapter 10, we saw a close connection between competitive, price-taking equilibria and Pareto optimality (or, Pareto efficiency).¹ The first welfare theorem tells us that competitive equilibria are necessarily Pareto optimal. From the second welfare theorem, we know that under suitable convexity hypotheses, any Pareto optimal allocation can be achieved as a competitive allocation after an appropriate lump-sum redistribution of wealth. Under the assumptions of these theorems, the possibilities for welfare-enhancing intervention in the marketplace are strictly limited to the carrying out of wealth transfers for the purposes of achieving distributional aims.

With this chapter, we begin our study of *market failures*: situations in which some of the assumptions of the welfare theorems do *not* hold and in which, as a consequence, market equilibria cannot be relied on to yield Pareto optimal outcomes. In this chapter, we study two types of market failure, known as *externalities* and *public goods*.

In Chapter 10, we assumed that the preferences of a consumer were defined solely over the set of goods that she might herself decide to consume. Similarly, the production of a firm depended only on its own input choices. In reality, however, a consumer or firm may in some circumstances be directly affected by the actions of other agents in the economy; that is, there may be *external effects* from the activities of other consumers or firms. For example, the consumption by consumer i 's neighbor of loud music at three in the morning may prevent her from sleeping. Likewise, a fishery's catch may be impaired by the discharges of an upstream chemical plant. Incorporating these concerns into our preference and technology formalism is, in principle, a simple matter: We need only define an agent's preferences or production set over both her own actions and those of the agent creating the external effect. But the effect on market equilibrium is significant: In general, when external effects are present, competitive equilibria are not Pareto optimal.

Public goods, as the name suggests, are commodities that have an inherently "public" character, in that consumption of a unit of the good by one agent does not preclude its consumption by another. Examples abound: Roadways, national defense,

1. See also Chapter 16.

flood-control projects, and knowledge all share this characteristic. The private provision of public goods generates a special type of externality: if one individual provides a unit of a public good, all individuals benefit. As a result, private provision of public goods is typically Pareto inefficient.

We begin our investigation of externalities and public goods in Section 11.B by considering the simplest possible externality: one that involves only two agents in the economy, where one of the agents engages in an activity that directly affects the other. In this setting, we illustrate the inefficiency of competitive equilibria when an externality is present. We then go on to consider three traditional solutions to this problem: quotas, taxes, and the fostering of decentralized bargaining over the extent of the externality. The last of these possibilities also suggests a connection between the presence of externalities and the nonexistence of certain commodity markets, a topic that we explore in some detail.

In Section 11.C, we study public goods. We first derive a condition that characterizes the optimal level of a public good and we then illustrate the inefficiency resulting from private provision. This Pareto inefficiency can be seen as arising from an externality among the consumers of the good, which in this context is known as the *free-rider problem*. We also discuss possible solutions to this free-rider problem. Both quantity-based intervention (here, direct governmental provision) and price-based intervention (taxes and subsidies) can, in principle, correct it. In contrast, decentralized bargaining and competitive market-based solutions are unlikely to be viable in the context of public goods.

In Section 11.D, we return to the analysis of externalities. We study cases in which many agents both produce and are affected by the externality. Multilateral externalities can be classified according to whether the externality is *depletable* (or *private* or *rivalrous*) or *nondepletable* (or *public* or *nonrivalrous*). We argue that market solutions are likely to work well in the former set of cases but poorly in the latter, where the externality possesses the characteristics of a public good (or bad). Indeed, this may well explain why most externalities that are regarded as serious social problems (e.g., water pollution, acid rain, congestion) take the form of nondepletable multilateral externalities.

In Section 11.E, we examine another problem that may arise in these settings: Individuals may have privately held information about the effects of externalities on their well-being. We see there that this type of informational asymmetry may confound both private and government efforts to achieve optimal outcomes.

In Appendix A, we study the connection between externalities and the presence of technological nonconvexities, and we examine the implications of these nonconvexities for our analysis.

The literature on externalities and public goods is voluminous. Useful introductions and further references to these subjects may be found in Baumol and Oates (1988) and Laffont (1988).

11.B A Simple Bilateral Externality

Surprisingly, perhaps, a fully satisfying definition of an externality has proved somewhat elusive. Nevertheless, informal Definition 11.B.1 provides a serviceable point of departure.

Definition 11.B.1: An *externality* is present whenever the well-being of a consumer or the production possibilities of a firm are directly affected by the actions of another agent in the economy.

Simple as Definition 11.B.1 sounds, it contains a subtle point that has been a source of some confusion. When we say “directly,” we mean to exclude any effects that are mediated by prices. That is, an externality is present if, say, a fishery’s productivity is affected by the emissions from a nearby oil refinery, but not simply because the fishery’s profitability is affected by the price of oil (which, in turn, is to some degree affected by the oil refinery’s output of oil). The latter type of effect [referred to as a *pecuniary externality* by Viner (1931)] is present in any competitive market but, as we saw in Chapter 10, creates no inefficiency. Indeed, with price-taking behavior, the market is precisely the mechanism that guarantees a Pareto optimal outcome. This suggests that the presence of an externality is not merely a technological phenomenon but also a function of the set of markets in existence. We return to this point later in the section.

In the remainder of this section, we explore the implications of external effects for competitive equilibria and public policy in the context of a very simple two-agent, partial equilibrium model. We consider two consumers, indexed by $i = 1, 2$, who constitute a small part of the overall economy. In line with this interpretation, we suppose that the actions of these consumers do not affect the prices $p \in \mathbb{R}^L$ of the L traded goods in the economy. At these prices, consumer i ’s wealth is w_i .

In contrast with the standard competitive model, however, we assume that each consumer has preferences not only over her consumption of the L traded goods ($x_{1i}, \dots, x_{Li}$) but also over some action $h \in \mathbb{R}_+$ taken by consumer 1. Thus, consumer i ’s (differentiable) utility function takes the form $u_i(x_{1i}, \dots, x_{Li}, h)$, and we assume that $\partial u_2(x_{12}, \dots, x_{L2}, h)/\partial h \neq 0$. Because consumer 1’s choice of h affects consumer 2’s well-being, it generates an externality. For example, the two consumers may live next door to each other, and h may be a measure of how loudly consumer 1 plays music. Or the consumers may live on a river, with consumer 1 further upstream. In this case, h could represent the amount of pollution put into the river by consumer 1; more pollution lowers consumer 2’s enjoyment of the river. We should hasten to add that external effects need not be detrimental to those affected by them. Action h could, for example, be consumer 1’s beautification of her property, which her neighbor, consumer 2, also gets to enjoy.²

In what follows, it will be convenient to define for each consumer i a derived utility function over the level of h , assuming optimal commodity purchases by consumer i at prices $p \in \mathbb{R}^L$ and wealth w_i :

$$v_i(p, w_i, h) = \max_{x_i \geq 0} u_i(x_i, h) \\ \text{s.t. } p \cdot x_i \leq w_i.$$

For expositional purposes, we shall also assume that the consumers’ utility functions

2. An externality favorable to the recipient is usually called a *positive externality*, and conversely for a *negative externality*.

take a quasilinear form with respect to a numeraire commodity (we comment below, in small type, on the simplifications afforded by this assumption). In this case, we can write the derived utility function $v_i(\cdot)$ as $v_i(p, w_i, h) = \phi_i(p, h) + w_i$.³ Since prices of the L traded goods are assumed to be unaffected by any of the changes we are considering, we shall suppress the price vector p and simply write $\phi_i(h)$. We assume that $\phi_i(\cdot)$ is twice differentiable with $\phi_i''(\cdot) < 0$. Be warned, however, that the concavity assumption is less innocent than it looks: see Appendix A for further discussion of this point.

Although we shall speak in terms of this consumer interpretation, everything we do here applies equally well to the case in which the two agents are firms (or, for that matter, one firm and one consumer). For example, we could consider a firm j that has a derived profit function $\pi_j(p, h)$ over h given prices p . Suppressing the price vector p , the firm's profit can be written as $\pi_j(h)$, which plays the same role as the function $\phi_i(h)$ in the analysis that follows.

Nonoptimality of the Competitive Outcome

Suppose that we are at a competitive equilibrium in which commodity prices are p . That is, at the equilibrium position, each of the two consumers maximizes her utility limited only by her wealth and the prices p of the traded goods. It must therefore be the case that consumer 1 chooses her level of $h \geq 0$ to maximize $\phi_1(h)$. Thus, the equilibrium level of h , h^* , satisfies the necessary and sufficient first-order condition

$$\phi_1'(h^*) \leq 0, \text{ with equality if } h^* > 0. \quad (11.B.1)$$

For an interior solution, we therefore have $\phi_1'(h^*) = 0$.

In contrast, in any Pareto optimal allocation, the optimal level of h , h° , must maximize the *joint surplus* of the two consumers, and so must solve⁴

$$\text{Max}_{h \geq 0} \phi_1(h) + \phi_2(h).$$

This problem gives us the necessary and sufficient first-order condition for h° of

$$\phi_1'(h^\circ) \leq -\phi_2'(h^\circ), \text{ with equality if } h^\circ > 0. \quad (11.B.2)$$

Hence, for an interior solution to the Pareto optimality problem, $\phi_1'(h^\circ) = -\phi_2'(h^\circ)$.

When external effects are present, so that $\phi_2'(h) \neq 0$ at all h , the equilibrium level of h is not optimal unless $h^\circ = h^* = 0$. Consider, for example, the case in which we have interior solutions, that is, where $(h^*, h^\circ) \gg 0$. If $\phi_2'(\cdot) < 0$, so that h generates

3. Indeed, suppose that $u_i(x_i, h) = g_i(x_{-1i}, h) + x_{1i}$, where x_{-1i} is consumer i 's consumption of traded goods other than good 1. Then, the consumer's Walrasian demand function for these $L - 1$ traded goods, $x_{-1i}(\cdot)$, is independent of her wealth, and $v_i(p, w_i, h) = g_i(x_{-1i}(p, h), h) - p \cdot x_{-1i}(p, h) + w_i$. Thus, denoting $\phi_i(p, h) = g_i(x_{-1i}(p, h), h) - p \cdot x_{-1i}(p, h)$, we have obtained the desired form.

4. Recall the reasoning of Sections 10.D and 10.E, or note that at any Pareto optimal allocation in which h° is the level of h and w_i is consumer i 's wealth level for $i = 1, 2$, it must be impossible to change h and reallocate wealth so as to make one consumer better off without making the other worse off. Thus, $(h^\circ, 0)$ must solve $\text{Max}_{h, T} \phi_1(h) + w_1 - T$ subject to $\phi_2(h) + w_2 + T \geq \bar{u}_2$, for some $\bar{u}_2$. Because the constraint holds with equality in any solution to this problem, substituting from the constraint for T in the objective function shows that h° must maximize the joint surplus of the two consumers $\phi_1(h) + \phi_2(h)$.

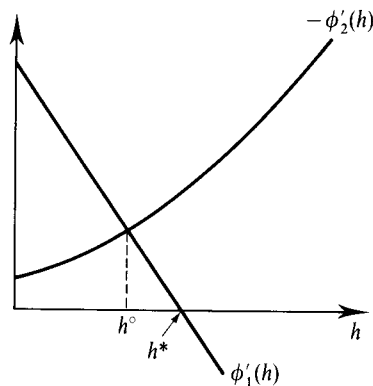


Figure 11.B.1
The equilibrium (h^*) and optimal (h°) levels of a negative externality.

a negative externality, then we have $\phi'_1(h^\circ) = -\phi'_2(h^\circ) > 0$; because $\phi'_1(\cdot)$ is decreasing and $\phi'_1(h^*) = 0$, this implies that $h^* > h^\circ$. In contrast, when $\phi'_2(\cdot) > 0$, h represents a positive externality, and $\phi'_1(h^\circ) = -\phi'_2(h^\circ) < 0$ implies that $h^* < h^\circ$.

Figure 11.B.1 depicts the solution for a case in which h constitutes a negative external effect, so that $\phi'_2(h) < 0$ at all h . In the figure, we graph $\phi'_1(\cdot)$ and $-\phi'_2(\cdot)$. The competitive equilibrium level of the externality h^* occurs at the point where the graph of $\phi'_1(\cdot)$ crosses the horizontal axis. In contrast, the optimal externality level h° corresponds to the point of intersection between the graphs of the two functions.

Note that optimality does not usually entail the complete elimination of a negative externality. Rather, the externality's level is adjusted to the point where the marginal benefit to consumer 1 of an additional unit of the externality-generating activity, $\phi'_1(h^\circ)$, equals its marginal cost to consumer 2, $-\phi'_2(h^\circ)$.

In the current example, quasilinear utilities lead the optimal level of the externality to be independent of the consumers' wealth levels. In the absence of quasilinearity, however, wealth effects for the consumption of the externality make its optimal level depend on the consumers' wealths. See Exercise 11.B.2 for an illustration. Note, however, that when the agents under consideration are firms, wealth effects are always absent.

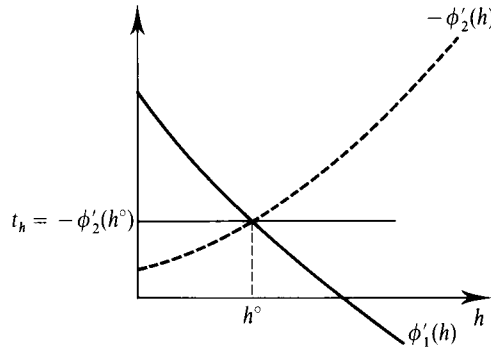
Traditional Solutions to the Externality Problem

Having identified the inefficiency of the competitive market outcome in the presence of an externality, we now consider three possible solutions to the problem. We first look at government-implemented quotas and taxes, and then analyze the possibility that an efficient outcome can be achieved in a much less intrusive manner by simply fostering bargaining between the consumers over the extent of the externality.

Quotas and taxes

To fix ideas, suppose that h generates a negative external effect, so that $h^\circ < h^*$. The most direct sort of government intervention to achieve efficiency is the direct control of the externality-generating activity itself. The government can simply mandate that h be no larger than h° , its optimal level. With this constraint, consumer 1 will indeed fix the level of the externality at h° .

A second option is for the government to attempt to restore optimality by imposing a tax on the externality-generating activity. This solution is known as

**Figure 11.B.2**

The optimality-restoring Pigouvian tax.

Pigouvian taxation, after Pigou (1932). To this effect, suppose that consumer 1 is made to pay a tax of t_h per unit of h . It is then not difficult to see that a tax of

$$t_h = -\phi'_2(h^\circ) > 0$$

will implement the optimal level of the externality. Indeed, consumer 1 will then choose the level of h that solves

$$\text{Max}_{h \geq 0} \phi_1(h) - t_h h, \quad (11.B.3)$$

which has the necessary and sufficient first-order condition

$$\phi'_1(h) \leq t_h, \quad \text{with equality if } h > 0. \quad (11.B.4)$$

Given $t_h = -\phi'_2(h^\circ)$, $h = h^\circ$ satisfies condition (11.B.4) [recall that h° is defined by the condition: $\phi'_1(h^\circ) \leq -\phi'_2(h^\circ)$, with equality if $h^\circ > 0$]. Moreover, given $\phi''_1(\cdot) < 0$, h° must be the unique solution to problem (11.B.3). Figure 11.B.2 illustrates this solution for a case in which $h^\circ > 0$.

Note that the optimality-restoring tax is exactly equal to the *marginal externality* at the optimal solution.⁵ That is, it is exactly equal to the amount that consumer 2 would be willing to pay to reduce h slightly from its optimal level h° . When faced with this tax, consumer 1 is effectively led to carry out an individual cost-benefit computation that *internalizes* the externality that she imposes on consumer 2.

The principles for the case of a positive externality are exactly the same, only now when we set $t_h = -\phi'_2(h^\circ) < 0$, t_h takes the form of a per-unit *subsidy* (i.e., consumer 1 *receives* a payment for each unit of the externality she generates).

Several additional points are worth noting about this Pigouvian solution. First, we can actually achieve optimality either by taxing the externality or by subsidizing its reduction. Consider, for example, the case of a negative externality. Suppose the government pays a subsidy of $s_h = -\phi'_2(h^\circ) > 0$ for every unit that consumer 1's choice of h is below h^* , its level in the competitive equilibrium. If so, then consumer 1 will maximize $\phi_1(h) + s_h(h^* - h) = \phi_1(h) - t_h h + t_h h^*$. But this is equivalent to a tax of t_h per unit on h combined with a lump-sum payment of $t_h h^*$. Hence, a subsidy for the reduction of the externality combined with a lump-sum transfer can exactly replicate the outcome of the tax.

Second, a point implicit in the derivation above is that, in general, it is essential

5. In the case where $h^\circ = 0$, any tax greater than $-\phi'_2(0)$ also implements the optimal outcome.

to tax the externality-producing activity directly. For instance, suppose that, in the example of consumer 1 playing loud music, we tax purchases of music equipment instead of taxing the playing of loud music itself. In general, this will not restore optimality. Consumer 1 will be led to lower her consumption of music equipment (perhaps she will purchase only a CD player, rather than a CD player and a tape player) but may nevertheless play whatever equipment she does purchase too loudly. A common example of this sort arises when a firm pollutes in the process of producing output. A tax on its output leads the firm to reduce its output level but may not have any effect (or, more generally, may have too little effect) on its pollution emissions. Taxing output achieves optimality only in the special case in which emissions bear a fixed monotonic relationship to the level of output. In this special case, emissions can be measured by the level of output, and a tax on output is essentially equivalent to a tax on emissions. (See Exercise 11.B.5 for an illustration.)

Third, note that the tax/subsidy and the quota approaches are equally effective in achieving an optimal outcome. However, the government must have a great deal of information about the benefits and costs of the externality for the two consumers to set the optimal levels of either the quota or the tax. In Section 11.E we will see that when the government does not possess this information the two approaches typically are not equivalent.

Fostering bargaining over externalities: enforceable property rights

Another approach to the externality problem aims at a less intrusive form of intervention, merely seeking to insure that conditions are met for the parties to themselves reach an optimal agreement on the level of the externality.

Suppose that we establish enforceable property rights with regard to the externality-generating activity. Say, for example, that we assign the right to an “externality-free” environment to consumer 2. In this case, consumer 1 is unable to engage in the externality-producing activity without consumer 2’s permission. For simplicity, imagine that the bargaining between the parties takes a form in which consumer 2 makes consumer 1 a take-it-or-leave-it offer, demanding a payment of T in return for permission to generate externality level h .⁶ Consumer 1 will agree to this demand if and only if she will be at least as well off as she would be by rejecting it, that is, if and only if $\phi_1(h) - T \geq \phi_1(0)$. Hence, consumer 2 will choose her offer (h, T) to solve

$$\begin{aligned} \text{Max}_{h \geq 0, T} \quad & \phi_2(h) + T \\ \text{s.t.} \quad & \phi_1(h) - T \geq \phi_1(0). \end{aligned}$$

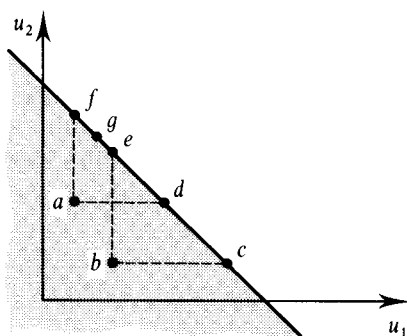
Since the constraint is binding in any solution to this problem, $T = \phi_1(h) - \phi_1(0)$. Therefore, consumer 2’s optimal offer involves the level of h that solves

$$\text{Max}_{h \geq 0} \quad \phi_2(h) + \phi_1(h) - \phi_1(0). \quad (11.B.5)$$

But this is precisely h° , the socially optimal level.

Note, moreover, that the precise allocation of these rights between the two

6. Either of the bargaining processes discussed in Appendix A of Chapter 9 would yield the same conclusions.

**Figure 11.B.3**

The final distribution of utilities under different property rights institutions and different bargaining procedures.

consumers is inessential to the achievement of optimality. Suppose, for example, that consumer 1 instead has the right to generate as much of the externality as she wants. In this case, in the absence of any agreement, consumer 1 will generate externality level h^* . Now consumer 2 will need to offer a $T < 0$ (i.e., to pay consumer 1) to have $h < h^*$. In particular, consumer 1 will agree to externality level h if and only if $\phi_1(h) - T \geq \phi_1(h^*)$. As a consequence, consumer 2 will offer to set h at the level that solves $\text{Max}_{h \geq 0} (\phi_2(h) + \phi_1(h) - \phi_1(h^*))$. Once again, the optimal externality level h° results. The allocation of rights affects only the final wealth of the two consumers by altering the payment made by consumer 1 to consumer 2. In the first case, consumer 1 pays $\phi_1(h^\circ) - \phi_1(0) > 0$ to be allowed to set $h^\circ > 0$; in the second, she “pays” $\phi_1(h^\circ) - \phi_1(h^*) < 0$ in return for setting $h^\circ < h^*$.

We have here an instance of what is known as the *Coase theorem* [for Coase (1960)]: If trade of the externality can occur, then bargaining will lead to an efficient outcome no matter how property rights are allocated.

All this is illustrated in Figure 11.B.3, in which we represent the utility possibility set for the two consumers. Every point in the boundary of this set corresponds to an allocation with externality level h° . The points a and b correspond to the utility levels arising, respectively, from externality levels 0 and h^* in the absence of any transfers. They constitute the initial situation after the assignment of property rights (to consumers 2 and 1, respectively) but before bargaining. In the particular bargaining procedure we have adopted (which gives the power to make a take-it-or-leave-it offer to consumer 2), the utility levels after bargaining are points f and e , respectively. If the bargaining power (i.e., the power to make the take-it-or-leave-it offer) had been instead in the hands of consumer 1, the post-bargaining utility levels would have been points d and c , respectively. Other bargaining procedures (such as the ones studied in Appendix A of Chapter 9) may yield other points in the segments $[f, d]$ and $[e, c]$, respectively.

Note that the existence of both well-defined and enforceable property rights is essential for this type of bargaining to occur. If property rights are not well defined, it will be unclear whether consumer 1 must gain consumer 2's permission to generate the externality. If property rights cannot be enforced (perhaps the level of h is not easily measured), then consumer 1 has no need to purchase the right to engage in the externality-generating activity from consumer 2. For this reason, proponents of this type of approach focus on the absence of these legal institutions as a central impediment to optimality.

This solution to the externality problem has a significant advantage over the tax and quota schemes in terms of the level of knowledge required of the government. The consumers must know each other's preferences, but the government need not. We should emphasize, however, that for bargaining over the externality to lead to efficiency, it is important that the consumers know this information. In Section 11.E, we will see that when the agents are to some extent ignorant of each others' preferences, bargaining need *not* lead to an efficient outcome.

Two further points regarding these three types of solutions to the externality problem are worthy of note. First, in the case in which the two agents are firms, one form that an efficient bargain might take is the sale of one of the firms to the other. The resulting merged firm would then fully internalize the externality in the process of maximizing its profits.⁷

Second, note that all three approaches require that the externality-generating activity be measureable. This is not a trivial requirement; in many cases, such measurement may be either technologically infeasible or very costly (consider the cost of measuring air pollution or noise). A proper computation of costs and benefits should take these costs into account. If measurement is very costly, then it may be optimal to simply allow the externality to persist.

Externalities and Missing Markets

The observation that bargaining can generate an optimal outcome suggests a connection between externalities and missing markets. After all, a market system can be viewed as a particular type of trading procedure.

Suppose that property rights are well defined and enforceable and that a competitive market for the right to engage in the externality-generating activity exists. For simplicity, we assume that consumer 2 has the right to an externality-free environment. Let p_h denote the price of the right to engage in one unit of the activity. In choosing how many of these rights to purchase, say h_1 , consumer 1 will solve

$$\text{Max}_{h_1 \geq 0} \phi_1(h_1) - p_h h_1,$$

which has the first-order condition

$$\phi'_1(h_1) \leq p_h, \quad \text{with equality if } h_1 > 0. \quad (11.B.6)$$

In deciding how many rights to sell, h_2 , consumer 2 will solve

$$\text{Max}_{h_2 \geq 0} \phi_2(h_2) + p_h h_2,$$

which has the first-order condition

$$\phi'_2(h_2) \leq -p_h, \quad \text{with equality if } h_2 > 0. \quad (11.B.7)$$

7. Note, however, that this conclusion presumes that the owner of a firm has full control over all its functions. In more complicated (but realistic) settings in which this is not true, say because owners must hire managers whose actions cannot be perfectly controlled, the results of a merger and of an agreement over the level of the externality need not be the same. Chapters 14 and 23 provide an introduction to the topic of incentive design. See Holmstrom and Tirole (1989) for a discussion of these issues in the theory of the firm.

In a competitive equilibrium, the market for these rights must clear; that is, we must have $h_1 = h_2$. Hence, (11.B.6) and (11.B.7) imply that the level of rights traded in this competitive rights market, say h^{**} , satisfies

$$\phi'_1(h^{**}) \leq -\phi'_2(h^{**}), \quad \text{with equality if } h^{**} > 0.$$

Comparing this expression with (11.B.2), we see that h^{**} equals the optimal level h° . The equilibrium price of the externality is $p_h^* = \phi'_1(h^\circ) = -\phi'_2(h^\circ)$.

Consumer 1 and 2's equilibrium utilities are then $\phi_1(h^\circ) - p_h^* h^\circ$ and $\phi_2(h^\circ) + p_h^* h^\circ$, respectively. The market therefore works as a particular bargaining procedure for splitting the gains from trade; for example, point g in Figure 11.B.3 could represent the utilities in the competitive equilibrium.

We see that if a competitive market exists for the externality, then optimality results. Thus, externalities can be seen as being inherently tied to the absence of certain competitive markets, a point originally noted by Meade (1952) and substantially extended by Arrow (1969). Indeed, recall that our original definition of an externality, Definition 11.B.1, explicitly required that an action chosen by one agent must *directly* affect the well-being or production capabilities of another. Once a market exists for an externality, however, each consumer decides for herself how much of the externality to consume at the going prices.

Unfortunately, the idea of a competitive market for the externality in the present example is rather unrealistic; in a market with only one seller and one buyer, price taking would be unlikely.⁸ However, most important externalities are produced and felt by *many* agents. Thus, we might hope that in these multilateral settings, price taking would be a more reasonable assumption and, as a result, that a competitive market for the externality would lead to an efficient outcome. In Section 11.D, where we study multilateral externalities, we see that the correctness of this conclusion depends on whether the externality is "private" or "public" in nature. Before coming to this, however, we first study the nature of public goods.

11.C Public Goods

In this section, we study commodities that, in contrast with those considered so far, have a feature of "publicness" to their consumption. These commodities are known as *public goods*.

Definition 11.C.1: A *public good* is a commodity for which use of a unit of the good by one agent does not preclude its use by other agents.

Put somewhat differently, public goods possess the feature that they are *nondepletable*: Consumption by one individual does not affect the supply available for other individuals. Knowledge provides a good illustration. The use of a piece of knowledge for one purpose does not preclude its use for others. In contrast, the commodities studied up to this point have been assumed to be of a *private*, or *depletable*, nature;

8. For that matter, the idea that the externality rights are all sold at the same price lacks justification here, because there is no natural unit of measurement for the externality.

that is, for each additional unit consumed by individual i , there is one unit less available for individuals $j \neq i$.⁹

A distinction can also be made according to whether *exclusion* of an individual from the benefits of a public good is possible. Every private good is automatically excludable, but public goods may or may not be. The patent system, for example, is a mechanism for excluding individuals (although imperfectly) from the use of knowledge developed by others. On the other hand, it might be technologically impossible, or at the least very costly, to exclude some consumers from the benefits of national defense or of a project to improve air quality. For simplicity, our discussion here will focus primarily on the case in which exclusion is not possible.

Note that a public “good” need not necessarily be desirable; that is, we may have public *bads* (e.g., foul air). In this case, we should read the phrase “does not preclude” in Definition 11.C.1 to mean “does not decrease.”

Conditions for Pareto Optimality

Consider a setting with I consumers and one public good, in addition to L traded goods of the usual, private, kind. We again adopt a partial equilibrium perspective by assuming that the quantity of the public good has no effect on the prices of the L traded goods and that each consumer's utility function is quasilinear with respect to the same numeraire, traded commodity. As in Section 11.B, we can therefore define, for each consumer i , a derived utility function over the level of the public good. Letting x denote the quantity of the public good, we denote consumer i 's utility from the public good by $\phi_i(x)$. We assume that this function is twice differentiable, with $\phi_i''(x) < 0$ at all $x \geq 0$. Note that precisely because we are dealing with a public good, the argument x does not have an i subscript.

The cost of supplying q units of the public good is $c(q)$. We assume that $c(\cdot)$ is twice differentiable, with $c''(q) > 0$ at all $q \geq 0$.

To describe the case of a desirable public good whose production is costly, we take $\phi_i'(\cdot) > 0$ for all i and $c'(\cdot) > 0$. Except where otherwise noted, however, the analysis applies equally well to the case of a public bad whose reduction is costly, where $\phi_i'(\cdot) < 0$ for all i and $c'(\cdot) < 0$.

In this quasilinear model, any Pareto optimal allocation must maximize aggregate surplus (see Section 10.D) and therefore must involve a level of the public good that solves

$$\text{Max}_{q \geq 0} \sum_{i=1}^I \phi_i(q) - c(q).$$

The necessary and sufficient first-order condition for the optimal quantity q° is then

$$\sum_{i=1}^I \phi_i'(q^\circ) \leq c'(q^\circ), \quad \text{with equality if } q^\circ > 0. \quad (11.C.1)$$

Condition (11.C.1) is the classic optimality condition for a public good first derived by Samuelson (1954; 1955). (Here it is specialized to the partial equilibrium setting;

9. Intermediate cases are also possible in which the consumption of the good by one individual affects to some degree its availability to others. A classic example is the presence of congestion effects. For this reason, goods for which there is no depletable whatsoever are sometimes referred to as *pure* public goods.

see Section 16.G for a more general treatment.) At an interior solution, we have $\sum_i \phi'_i(q^*) = c'(q^*)$, so that at the optimal level of the public good *the sum of consumers' marginal benefits from the public good is set equal to its marginal cost*. This condition should be contrasted with conditions (10.D.3) to (10.D.5) for a private good, where *each* consumer's marginal benefit from the good is equated to its marginal cost.

Inefficiency of Private Provision of Public Goods

Consider the circumstance in which the public good is provided by means of private purchases by consumers. We imagine that a market exists for the public good and that each consumer i chooses how much of the public good to buy, denoted by $x_i \geq 0$, taking as given its market price p . The total amount of the public good purchased by consumers is then $x = \sum_i x_i$. Formally, we treat the supply side as consisting of a single profit-maximizing firm with cost function $c(\cdot)$ that chooses its production level taking the market price as given. Note, however, that by the analysis of Section 5.E, we can actually think of the supply behavior of this firm as representing the industry supply of J price-taking firms whose aggregate cost function is $c(\cdot)$.

At a competitive equilibrium involving price p^* , each consumer i 's purchase of the public good x_i^* must maximize her utility and so must solve

$$\text{Max}_{x_i \geq 0} \phi_i(x_i + \sum_{k \neq i} x_k^*) - p^* x_i. \quad (11.C.2)$$

In determining her optimal purchases, consumer i takes as given the amount of the private good being purchased by each other consumer (as in the Nash equilibrium concept studied in Section 8.D). Consumer i 's purchases x_i^* must therefore satisfy the necessary and sufficient first-order condition

$$\phi'_i(x_i^* + \sum_{k \neq i} x_k^*) \leq p^*, \quad \text{with equality if } x_i^* > 0.$$

Letting $x^* = \sum_i x_i^*$ denote the equilibrium level of the public good, for each consumer i we must therefore have

$$\phi'_i(x^*) \leq p^*, \quad \text{with equality if } x_i^* > 0. \quad (11.C.3)$$

The firm's supply q^* , on the other hand, must solve $\text{Max}_{q \geq 0} (p^* q - c(q))$ and therefore must satisfy the standard necessary and sufficient first-order condition

$$p^* \leq c'(q^*), \quad \text{with equality if } q^* > 0. \quad (11.C.4)$$

At a competitive equilibrium, $q^* = x^*$. Thus, letting $\delta_i = 1$ if $x_i^* > 0$ and $\delta_i = 0$ if $x_i^* = 0$, (11.C.3) and (11.C.4) tell us that $\sum_i \delta_i [\phi'_i(q^*) - c'(q^*)] = 0$. Recalling that $\phi'_i(\cdot) > 0$ and $c'(\cdot) > 0$, this implies that whenever $I > 1$ and $q^* > 0$ (so that $\delta_i = 1$ for some i) we have

$$\sum_{i=1}^I \phi'_i(q^*) > c'(q^*). \quad (11.C.5)$$

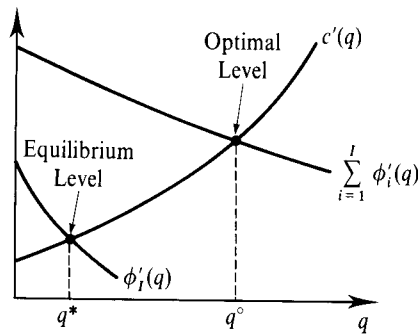


Figure 11.C.1

Private provision leads to an insufficient level of a desirable public good.

Comparing (11.C.5) with (11.C.1), we see that whenever $q^\circ > 0$ and $I > 1$, the level of the public good provided is too low; that is, $q^* < q^\circ$.¹⁰

The cause of this inefficiency can be understood in terms of our discussion of externalities in Section 11.B. Here each consumer's purchase of the public good provides a direct benefit not only to the consumer herself but also to every other consumer. Hence, private provision creates a situation in which externalities are present. The failure of each consumer to consider the benefits for others of her public good provision is often referred to as the *free-rider problem*: Each consumer has an incentive to enjoy the benefits of the public good provided by others while providing it insufficiently herself.

In fact, in the present model, the free-rider problem takes a very stark form. To see this most simply, suppose that we can order the consumers according to their marginal benefits, in the sense that $\phi'_1(x) < \dots < \phi'_I(x)$ at all $x \geq 0$. Then condition (11.C.3) can hold with equality only for a *single* consumer and, moreover, this must be the consumer labeled I . Therefore, only the consumer who derives the largest (marginal) benefit from the public good will provide it; all others will set their purchases equal to zero in the equilibrium. The equilibrium level of the public good is then the level q^* that satisfies $\phi'_I(q^*) = c'(q^*)$. Figure 11.C.1 depicts both this equilibrium and the Pareto optimal level. Note that the curve representing $\sum_i \phi'_i(q)$ geometrically corresponds to a *vertical* summation of the individual curves representing $\phi_i(q)$ for $i = 1, \dots, I$ (whereas in the case of a private good, the market demand curve is identified by adding the individual demand curves *horizontally*).

The inefficiency of private provision is often remedied by governmental intervention in the provision of public goods. Just as with externalities, this can happen not only through quantity-based intervention (such as direct governmental provision) but also through “price-based” intervention in the form of taxes or subsidies. For example, suppose that there are two consumers with benefit functions $\phi_1(x_1 + x_2)$ and $\phi_2(x_1 + x_2)$, where x_i is the amount of the public good purchased by consumer i , and that $q^\circ > 0$. By analogy with the analysis in Section 11.B, a subsidy to each consumer i per unit purchased of $s_i = \phi'_{-i}(q^\circ)$ [or, equivalently, a tax of $-\phi'_{-i}(q^\circ)$ per unit that consumer i 's purchases of the public good fall below some specified

10. The conclusion follows immediately if $q^* = 0$. So suppose instead that $q^* > 0$. Then since $\sum_i \phi'_i(q^*) - c'(q^*) > 0$ and $\sum_i \phi'_i(\cdot) - c'(\cdot)$ is decreasing, any solution to (11.C.1) must have a larger value than q^* . Note that, in contrast, if we are dealing with a public bad, so that $\phi'_i(\cdot) < 0$ and $c'(\cdot) < 0$, then the inequalities reverse and $q^\circ < q^*$.

level] faces each consumer with the marginal external effect of her actions and so generates an optimal level of public good provision by consumer i . Formally, if $(\tilde{x}_1, \tilde{x}_2)$ are the competitive equilibrium levels of the public good purchased by the two consumers given these subsidies, and if $\tilde{p}$ is the equilibrium price, then consumer i 's purchases of the public good, $\tilde{x}_i$, must solve $\text{Max}_{x_i \geq 0} \phi_i(x_i + \tilde{x}_j) + s_i x_i - \tilde{p} x_i$, and so $\tilde{x}_i$ must satisfy the necessary and sufficient first-order condition

$$\phi'_i(\tilde{x}_1 + \tilde{x}_2) + s_i \leq \tilde{p}, \text{ with equality if } \tilde{x}_i > 0.$$

Substituting for s_i , and using both condition (11.C.4) and the market-clearing condition that $\tilde{x}_1 + \tilde{x}_2 = \tilde{q}$, we conclude that $\tilde{q}$ is the total amount of the public good in the competitive equilibrium given these subsidies if and only if

$$\phi'_i(\tilde{q}) + \phi'_{-i}(q^\circ) \leq c'(\tilde{q}),$$

with equality for some i if $\tilde{q} > 0$. Recalling (11.C.1) we see that $\tilde{q} = q^\circ$. (Exercise 11.C.1 asks you to extend this argument to the case where $I > 2$; formally, we then have a multilateral externality of the sort studied in Section 11.D.)

Note that both optimal direct public provision and this subsidy scheme require that the government know the benefits derived by consumers from the public good (i.e., their willingness to pay in terms of private goods). In Section 11.E, we study the case in which this is not so.

Lindahl Equilibria

Although private provision of the sort studied above results in an inefficient level of the public good, there is *in principle* a market institution that can achieve optimality. Suppose that, for each consumer i , we have a market for the public good "as experienced by consumer i ." That is, we think of each consumer's consumption of the public good as a distinct commodity with its own market. We denote the price of this personalized good by p_i . Note that p_i may differ across consumers. Suppose also that, given the equilibrium price p_i^{**} , each consumer i sees herself as deciding on the *total amount of the public good she will consume*, x_i , so as to solve

$$\text{Max}_{x_i \geq 0} \phi_i(x_i) - p_i^{**} x_i.$$

Her equilibrium consumption level x_i^{**} must therefore satisfy the necessary and sufficient first-order condition

$$\phi'_i(x_i^{**}) \leq p_i^{**}, \text{ with equality if } x_i^{**} > 0. \quad (11.C.6)$$

The firm is now viewed as producing a bundle of I goods with a fixed-proportions technology (i.e., the level of production of each personalized good is necessarily the same). Thus, the firm solves

$$\text{Max}_{q \geq 0} \left(\sum_{i=1}^I p_i^{**} q \right) - c(q).$$

The firm's equilibrium level of output q^{**} therefore satisfies the necessary and

sufficient first-order condition

$$\sum_{i=1}^I p_i^{**} \leq c'(q^{**}), \quad \text{with equality if } q^{**} > 0. \quad (11.C.7)$$

Together, (11.C.6), (11.C.7), and the market-clearing condition that $x_i^{**} = q^{**}$ for all i imply that

$$\sum_{i=1}^I \phi'_i(q^{**}) \leq c'(q^{**}), \quad \text{with equality if } q^{**} > 0. \quad (11.C.8)$$

Comparing (11.C.8) with (11.C.1), we see that the equilibrium level of the public good consumed by each consumer is exactly the efficient level: $q^{**} = q^{\circ}$.

This type of equilibrium in personalized markets for the public good is known as a *Lindahl equilibrium*, after Lindahl (1919). [See also Milleron (1972) for a further discussion.] To understand why we obtain efficiency, note that once we have defined personalized markets for the public good, each consumer, taking the price in her personalized market as given, fully determines her own level of consumption of the public good; externalities are eliminated.

Yet, despite the attractive properties of Lindahl equilibria, their realism is questionable. Note, first, that the ability to exclude a consumer from use of the public good is essential if this equilibrium concept is to make sense; otherwise a consumer would have no reason to believe that in the absence of making any purchases of the public good she would get to consume none of it.¹¹ Moreover, even if exclusion is possible, these are markets with only a single agent on the demand side. As a result, price-taking behavior of the sort presumed is unlikely to occur.

The idea that inefficiencies can in principle be corrected by introducing the right kind of markets, encountered here and in Section 11.B, is a very general one. In particular cases, however, this “solution” may or may not be a realistic possibility. We encounter this issue again in our study of multilateral externalities in Section 11.D. As we shall see, these types of externalities often share many of the features of public goods.

11.D Multilateral Externalities

In most cases, externalities are felt and generated by numerous parties. This is particularly true of those externalities, such as industrial pollution, smog caused by automobile use, or congestion, that are widely considered to be “important” policy problems. In this section, we extend our analysis of externalities to these multilateral settings.

An important distinction can be made in the case of multilateral externalities according to whether the externality is *depletable* (or *private*, or *rivalrous*) or *nondepletable* (or *public*, or *nonrivalrous*). Depletable externalities have the feature that experience of the externality by one agent reduces the amount that will be felt by other agents. For example, if the externality takes the form of the dumping of garbage on people's property, if an additional unit of garbage is dumped on one

11. Thus, the possibility of exclusion can be important for efficient supply of the public good, even though the use of an exclusion technology is itself inefficient (a Pareto optimal allocation cannot involve any exclusion).

piece of property, that much less is left to be dumped on others.¹² Depletable externalities therefore share the characteristics of our usual (private) sort of commodity. In contrast, air pollution is a nondepletable externality; the amount of air pollution experienced by one agent is not affected by the fact that others are also experiencing it. Nondepletable externalities therefore have the characteristics of public goods (or bads).

In this section we argue that a decentralized market solution can be expected to work well for multilateral depletable externalities as long as well-defined and enforceable property rights can be created. In contrast, market-based solutions are unlikely to work in the nondepletable case, in parallel to our conclusions regarding public goods in Section 11.C.

We shall assume throughout this section that the agents who generate externalities are distinct from those who experience them. This simplification is inessential but eases the exposition and facilitates comparison with the previous sections (Exercise 11.D.2 asks you to consider the general case). For ease of reference, we assume here that the generators of the externality are firms and that those experiencing the externality are consumers. We also focus on the special, but central, case in which the externality generated by the firms is homogeneous (i.e., consumers are indifferent to the source of the externality). (Exercise 11.D.4 asks you to consider the case in which the source matters.)

We again adopt a partial equilibrium approach and assume that agents take as given the price vector p of L traded goods. There are J firms that generate the externality in the process of production. As discussed in Section 11.B, given price vector p , we can determine firm j 's derived profit function over the level of the externality it generates, $h_j \geq 0$, which we denote by $\pi_j(h_j)$. There are also I consumers, who have quasilinear utility functions with respect to a numeraire, traded commodity. Given price vector p , we denote by $\phi_i(\tilde{h}_i)$ consumer i 's derived utility function over the amount of the externality $\tilde{h}_i$ she experiences. We assume that $\pi_j(\cdot)$ and $\phi_i(\cdot)$ are twice differentiable with $\pi_j''(\cdot) < 0$ and $\phi_i''(\cdot) < 0$. To fix ideas, we shall focus on the case where $\phi_i'(\cdot) < 0$ for all i , so that we are dealing with a negative externality.

Depletable Externalities

We begin by examining the case of depletable externalities. As in Section 11.B, it is easy to see that the level of the (negative) externality is excessive at an unfettered competitive equilibrium. Indeed, at any competitive equilibrium, each firm j will wish to set the externality-generating activity at the level h_j^* satisfying the condition

$$\pi_j(h_j^*) \leq 0, \quad \text{with equality if } h_j^* > 0. \quad (11.D.1)$$

In contrast, any Pareto optimal allocation involves the levels $(\tilde{h}_1^\circ, \dots, \tilde{h}_I^\circ, h_1^\circ, \dots, h_J^\circ)$

12. A distinction can also be made as to whether a depletable externality is *allocable*. For example, acid rain is depletable in the sense that the total amount of chemicals put into the air will fall somewhere, but it is not readily allocable because where it falls is determined by weather patterns. Throughout this section, we take depletable externalities to be allocable. The analytical implications of nonallocable depletable externalities parallel those of nondepletable ones.

13. The firms are indifferent about which consumer is affected by their externality. Therefore, apart from the fact that $\sum_i \tilde{h}_i = \sum_j h_j^*$, the particular values of the individual $\tilde{h}_i$'s are indeterminate.

that solve¹⁴

$$\begin{aligned} \text{Max}_{\substack{(h_1, \dots, h_J) \geq 0 \\ (\tilde{h}_1, \dots, \tilde{h}_I) \geq 0}} \quad & \sum_{i=1}^I \phi_i(\tilde{h}_i) + \sum_{j=1}^J \pi_j(h_j) \\ \text{s.t.} \quad & \sum_{j=1}^J h_j = \sum_{i=1}^I \tilde{h}_i. \end{aligned} \quad (11.D.2)$$

The constraint in (11.D.2) reflects the depletable nature of the externality: If $\tilde{h}_i$ is increased by one unit, there is one unit less of the externality that needs to be experienced by others. Letting μ be the multiplier on this constraint, the necessary and sufficient first-order conditions to problem (11.D.2) are

$$\phi'_i(\tilde{h}_i^\circ) \leq \mu, \quad \text{with equality if } \tilde{h}_i^\circ > 0, \quad i = 1, \dots, I, \quad (11.D.3)$$

and

$$\mu \leq -\pi'_j(h_j^\circ), \quad \text{with equality if } h_j^\circ > 0, \quad j = 1, \dots, J. \quad (11.D.4)$$

Conditions (11.D.3) and (11.D.4), along with the constraint in problem (11.D.2), characterize the optimal levels of externality generation and consumption. Note that they exactly parallel the efficiency conditions for a private good derived in Chapter 10, conditions (10.D.3) to (10.D.5), where we interpret $-\pi'_j(\cdot)$ as firm j 's marginal cost of producing more of the externality. If well-defined and enforceable property rights can be specified over the externality, and if I and J are large numbers so that price taking is a reasonable hypothesis, then by analogy with the analysis of competitive markets for private goods in Chapter 10, a market for the externality can be expected to lead to the optimal levels of externality production and consumption in the depletable case.

Nondepletable Externalities

We now move to the case in which the externality is nondepletable. To be specific, assume that the level of the externality experienced by *each* consumer is $\sum_j h_j$, the total amount of the externality produced by the firms.

In an unfettered competitive equilibrium, each firm j 's externality generation h_j^* again satisfies condition (11.D.1). In contrast, any Pareto optimal allocation involves externality generation levels $(h_1^\circ, \dots, h_J^\circ)$ that solve

$$\text{Max}_{(h_1, \dots, h_J) \geq 0} \quad \sum_{i=1}^I \phi_i(\sum_j h_j) + \sum_{j=1}^J \pi_j(h_j). \quad (11.D.5)$$

This problem has necessary and sufficient first-order conditions for each firm j 's optimal level of externality generation, h_j° , of

$$\sum_{i=1}^I \phi'_i(\sum_j h_j^\circ) \leq -\pi'_j(h_j^\circ), \quad \text{with equality if } h_j^\circ > 0. \quad (11.D.6)$$

14. The objective function in (11.D.2) amounts to the usual difference between benefits and costs arising in the aggregate surplus measure. Note, to this effect, that $-\pi_j(\cdot)$ can be viewed as firm j 's cost function for producing the externality.

Condition (11.D.6) is exactly analogous to the optimality condition for a public good, condition (11.C.1), where $-\pi'_j(\cdot)$ is firm j 's marginal cost of externality production.¹⁵

By analogy with our discussion of private provision of public goods in Section 11.C, the introduction of a standard sort of market for the externality will *not* lead here, as it did in the bilateral case of Section 11.B, to an optimal outcome. The free-rider problem reappears, and the equilibrium level of the (negative) externality will exceed its optimal level. Instead, in the case of a multilateral nondepletable externality, a market-based solution would require personalized markets for the externality, as in the Lindahl equilibrium concept. However, all the problems with Lindahl equilibrium discussed in Section 11.C will similarly afflict these markets. As a result, purely market-based solutions, personalized or not, are unlikely to work in the case of a depletable externality.¹⁶

In contrast, given adequate information (a strong assumption!), the government *can* achieve optimality using quotas or taxes. With quotas, the government simply sets an upper bound on each firm j 's level of externality generation equal to its optimal level h_j° . On the other hand, as in Section 11.B, optimality-restoring taxes face each firm with the marginal social cost of their externality. Here the optimal tax is identical for each firm and is equal to $t_h = -\sum_i \phi'_i(\sum_j h_j^\circ)$ per unit of the externality generated. Given this tax, each firm j solves

$$\text{Max}_{h_j \geq 0} \quad \pi_j(h_j) - t_h h_j,$$

which has the necessary and sufficient first-order condition

$$\pi'_j(h_j) \leq t_h, \quad \text{with equality if } h_j > 0.$$

Given $t_h = -\sum_i \phi'_i(\sum_j h_j^\circ)$, firm j 's optimal choice is $h_j = h_j^\circ$.

A partial market-based approach that can achieve optimality with a nondepletable multilateral externality involves specification of a quota on the *total* level of the externality and distribution of that number of *tradeable externality permits* (each permit grants a firm the right to generate one unit of the externality). Suppose that $h^\circ = \sum_j h_j^\circ$ permits are given to the firms, with firm j receiving $\bar{h}_j$ of them. Let p_h^* denote the equilibrium price of these permits. Then each firm j 's demand for permits, h_j , solves $\text{Max}_{h_j \geq 0} (\pi_j(h_j) + p_h^*(\bar{h}_j - h_j))$ and so satisfies the necessary and sufficient first-order condition $\pi'_j(h_j) \leq p_h^*$, with equality if $h_j > 0$. In addition, market clearing in the permits market requires that $\sum_j h_j = h^\circ$. The competitive equilibrium in the market for permits then has price $p_h^* = -\sum_i \phi'_i(h^\circ)$ and each firm j using h_j° permits and so yields an optimal allocation. The advantage of this scheme relative to a strict quota method arises when the government has limited information about the $\pi_j(\cdot)$ functions and cannot tell which particular firms can efficiently bear the burden of externality reduction, although it has enough information, perhaps of a statistical sort, to allow the computation of the optimal aggregate level of the externality, h° .

15. Recall that the single firm's cost function $c(\cdot)$ in Section 11.C could be viewed as the aggregate cost function of J separate profit-maximizing firms. Were we to explicitly model these J firms in Section 11.C, the optimality conditions for public good production would take exactly the form in (11.D.6) with $c'_j(h_j^\circ)$ replacing $-\pi'_j(h_j^\circ)$.

16. The public nature of the externality leads to similar free-rider problems in any bargaining solution. (See Exercise 11.D.6 for an illustration.)

11.E Private Information and Second-Best Solutions

In practice, the degree to which an agent is affected by an externality or benefits from a public good will often be known only to her. The presence of *privately held* (or *asymmetrically held*) information can confound both centralized (e.g., quotas and taxes) and decentralized (e.g., bargaining) attempts to achieve optimality. In this section, we provide an introduction to these issues, focusing for the sake of specificity on the case of a bilateral externality such as that studied in Section 11.B. Following the convention adopted in Section 11.D, we shall assume here that the externality-generating agent is a firm and the affected agent is a consumer. (For a more general treatment of some of the topics covered in this section, see Chapter 23.)

Suppose, then, that we can write the consumer's derived utility function from externality level h (see Section 11.B for more on this construction) as $\phi(h, \eta)$, where $\eta \in \mathbb{R}$ is a parameter, to be called the consumer's *type*, that affects the consumer's costs from the externality. Similarly, we let $\pi(h, \theta)$ denote the firm's derived profit given its type $\theta \in \mathbb{R}$. The actual values of θ and η are *privately observed*: Only the consumer knows her type η , and only the firm observes its type θ . The ex ante likelihoods (probability distributions) of various values of θ and η are, however, publicly known. For convenience, we assume that θ and η are independently distributed. As previously, we assume that $\pi(h, \theta)$ and $\phi(h, \eta)$ are strictly concave in h for any given values of θ and η .

Decentralized Bargaining

Consider the decentralized approach to the externality problem first. In general, bargaining in the presence of bilateral asymmetric information will *not* lead to an efficient level of the externality. To see this, consider again the case in which the consumer has the right to an externality-free environment, and the simple bargaining process in which the consumer makes a take-it-or-leave-it offer to the firm. For simplicity, we assume that there are only two possible levels of the externality, 0 and $\bar{h} > 0$, and we focus on the case of a negative externality in which externality level $\bar{h}$, relative to the level 0, is detrimental for the consumer and beneficial for the firm (the analysis is readily applied to the case of a positive externality).

It is convenient to define $b(\theta) = \pi(\bar{h}, \theta) - \pi(0, \theta) > 0$ as the measure of the firm's benefit from the externality-generating activity when its type is θ . Similarly, we let $c(\eta) = \phi(0, \eta) - \phi(\bar{h}, \eta) > 0$ give the consumer's cost from externality level $\bar{h}$. In this simplified setting, the only aspects of the consumer's and firm's types that matter are the values of b and c that these types generate. Hence, we can focus directly on the various possible values of b and c that the two agents might have. Denote by $G(b)$ and $F(c)$ the distribution functions of these two variables induced by the underlying probability distributions of θ and η (note that, given the independence of θ and η , b and c are independent). For simplicity, we assume that these distributions have associated density functions $g(b)$ and $f(c)$, with $g(b) > 0$ and $f(c) > 0$ for all $b > 0$ and $c > 0$.

Since the consumer has the right to an externality-free environment, in the absence of any agreement with the firm she will always insist that the firm set $h = 0$ (recall that $c > 0$). However, in any arrangement that guarantees Pareto optimal outcomes for all values of b and c , the firm should be allowed to set $h = \bar{h}$ whenever $b > c$.

Now consider the amount that the consumer will demand from the firm when her cost is c in exchange for permission to engage in the externality-generating activity. Since the firm knows that the consumer will insist on $h = 0$ if there is no agreement, the firm will agree to pay the amount T if and only if $b \geq T$. Hence, the consumer knows that if she demands a payment of T , the probability that the firm will accept her offer equals the probability that $b \geq T$; that is, it is equal to $1 - G(T)$. Given her cost $c > 0$ (and assuming risk neutrality), the consumer optimally chooses the value of T she demands to solve

$$\max_T (1 - G(T))(T - c). \quad (11.E.1)$$

The objective function of problem (11.E.1) is the probability that the firm accepts the demand, multiplied by the net gain to the consumer when this happens ($T - c$). Under our assumptions, the objective function in (11.E.1) is strictly positive for all $T > c$ and equal to zero when $T = c$. Therefore, the solution, say T_c^* , is such that $T_c^* > c$. But this implies that this bargaining process must result in a strictly positive probability of an inefficient outcome, since whenever the firm's benefit b satisfies $c < b < T_c^*$, the firm will reject the consumer's offer, resulting in an externality level of zero, even though optimality requires that $h = \bar{h}$.^{17,18}

Quotas and Taxes

Just as decentralized bargaining will involve inefficiencies in the presence of privately held information, so too will the use of quotas and taxes. Moreover, as originally noted by Weitzman (1974), the presence of asymmetrically held information causes these two policy instruments to no longer be perfect substitutes for one another, as they were in the model of Section 11.B.¹⁹

To begin, note that given θ and η , the aggregate surplus resulting from externality level h (we return to a continuum of possible externality levels here) is $\phi(h, \eta) + \pi(h, \theta)$. Thus, the externality level that maximizes aggregate surplus depends in general on the realized values of (θ, η) . We denote this optimal value by the function $h^\circ(\theta, \eta)$. Figure 11.E.1 depicts this optimum value for two different pairs of parameters, (θ', η') and (θ'', η'') .

Suppose, first, that a quota level of $\hat{h}$ is fixed. The firm will then choose the level of the externality to solve

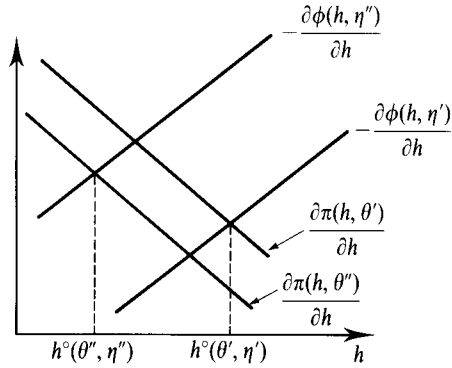
$$\begin{aligned} \max_{h \geq 0} \quad & \pi(h, \theta) \\ \text{s.t.} \quad & h \leq \hat{h}. \end{aligned}$$

Denote its optimal choice by $h^q(\hat{h}, \theta)$. The typical effect of the quota will be to make

17. Note the similarity between problem (11.E.1) and the monopolist's problem studied in Section 12.B. Here the consumer's inability to discriminate among firms of different types leads her optimal offer to be one that yields an inefficient outcome.

18. We could, of course, also consider the outcomes from other, perhaps more elaborate, bargaining procedures. In Chapter 23, however, we shall study a result due to Myerson and Satterthwaite (1983) that implies that *no* bargaining procedure can lead to an efficient outcome for all values of b and c in this setting.

19. The discussion that follows also has implications for the relative advantages of quantity-versus price-based control mechanisms in organizations.

**Figure 11.E.1**

The surplus-maximizing aggregate externality level for two different pairs of parameters, (θ', η') and (θ'', η'') .

the actual level of the externality much less sensitive to the values of θ and η than is required by optimality. The firm's externality level will be completely insensitive to η . Moreover, if the level of the quota $\hat{h}$ is such that $\partial\pi(\hat{h}, \theta)/\partial h > 0$ for all θ , we will have $h^q(\hat{h}, \theta) = \hat{h}$ for every θ . The loss in aggregate surplus arising under the quota for types (θ, η) is given by

$$\begin{aligned} \phi(h^q(\hat{h}, \theta), \eta) + \pi(h^q(\hat{h}, \theta), \theta) - \phi(h^\circ(\theta, \eta), \eta) - \pi(h^\circ(\theta, \eta), \theta) \\ = \int_{h^\circ(\theta, \eta)}^{h^q(\hat{h}, \theta)} \left(\frac{\partial\pi(h, \theta)}{\partial h} + \frac{\partial\phi(h, \eta)}{\partial h} \right) dh. \end{aligned}$$

This loss is represented by the shaded region in Figure 11.E.2 for a case in which the quota is set equal to $\hat{h} = h^\circ(\bar{\theta}, \bar{\eta})$, the externality level that maximizes social surplus when θ and η each take their mean values, $\bar{\theta}$ and $\bar{\eta}$ [the dashed lines in the figure are the graphs of $\partial\pi(h, \bar{\theta})/\partial h$ and $-\partial\phi(h, \bar{\eta})/\partial h$ and the solid lines are the graphs of $\partial\pi(h, \theta)/\partial h$ and $-\partial\phi(h, \eta)/\partial h$; note that in the case depicted, the firm wishes to produce the externality up to the allowed quota $\hat{h}$].

Consider next the use of a tax on the firm of t units of the numeraire per unit of the externality. For any given value of θ , the firm will then choose the level of externality to solve

$$\begin{aligned} \text{Max}_{h \geq 0} \quad & \pi(h, \theta) - th. \end{aligned}$$

Denote its optimal choice by $h^t(t, \theta)$. The loss in aggregate surplus from the tax relative to the optimal outcome for types (θ, η) is therefore given by

$$\begin{aligned} \phi(h^t(t, \theta), \eta) + \pi(h^t(t, \theta), \theta) - \phi(h^\circ(\theta, \eta), \eta) - \pi(h^\circ(\theta, \eta), \theta) \\ = \int_{h^\circ(\theta, \eta)}^{h^t(t, \theta)} \left(\frac{\partial\pi(h, \theta)}{\partial h} + \frac{\partial\phi(h, \eta)}{\partial h} \right) dh. \end{aligned}$$

Its value is depicted by the shaded region in Figure 11.E.3 for the same situation as in Figure 11.E.2, but now assuming that a tax is set at $t = -\partial\phi(h^\circ(\bar{\theta}, \bar{\eta}), \bar{\eta})/\partial h$, the value that results in the maximization of aggregate surplus when $(\theta, \eta) = (\bar{\theta}, \bar{\eta})$. Note that under a tax, as under a quota, the level of the externality is responsive to changes in the marginal benefits of the firm but not to changes in the marginal costs of the consumer.

Which of these instruments, quota or tax, performs better? The answer is that it depends. Imagine, for example, that η is a constant, say equal to $\bar{\eta}$. Then, for θ such that the benefits of the externality's use to the firm are high, a quota will typically miss the optimal externality level by not allowing the externality to increase above the quota level. On the other hand, because a fixed tax rate t does not reflect any

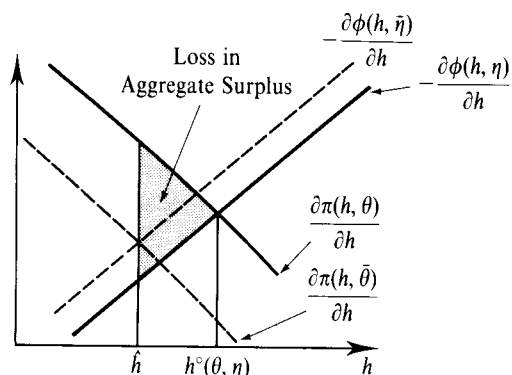


Figure 11.E.2 (left)

The loss in aggregate surplus under a quota for types (θ, η) .

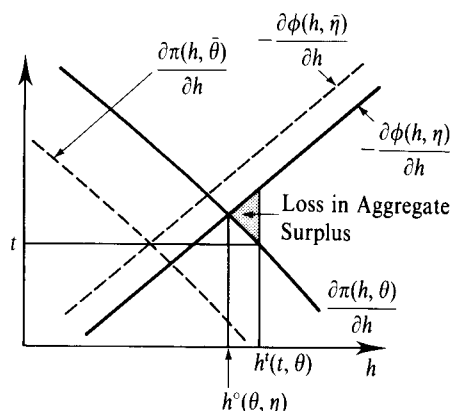


Figure 11.E.3 (right)

The loss in aggregate surplus under a tax for types (θ, η) .

increasing marginal costs of the externality to the consumer at higher externality levels, for such values of θ the tax may result in excess production of the externality.

Intuitively, when the optimal externality level varies little with θ , we expect a quota to be better. Figure 11.E.4(a), for example, depicts a case in which the marginal cost to the consumer of the externality is zero up to some point h^* and infinite thereafter. In this case, by setting a quota of $\hat{h} = h^*$, we can maximize aggregate surplus for any value of (θ, η) , but no tax can accomplish this. A tax would have to be very high to guarantee that with probability one the externality level fixed by the firm is not larger than h^* . But if so, the resulting externality level would be too low most of the time.

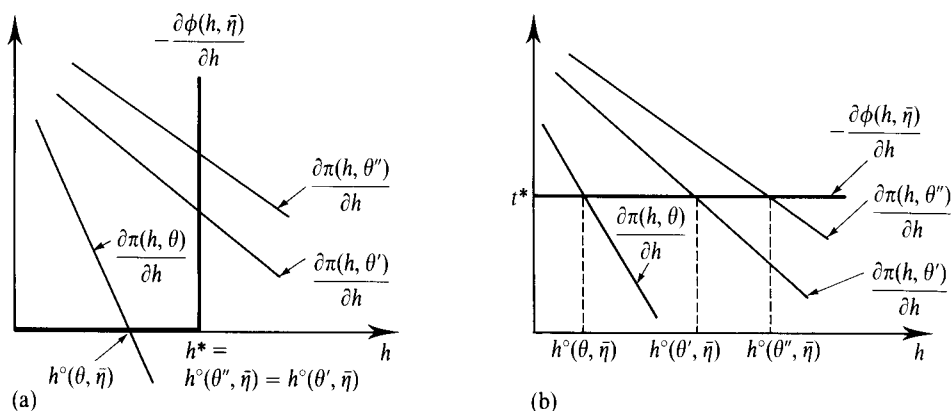
In contrast, in Figure 11.E.4(b) we depict a case in which the marginal cost to the consumer of the externality is independent of the level of h . In this case, a tax equal to this marginal cost ($t = t^*$) achieves the surplus-maximizing externality level for all (θ, η) , but no quota can do so.

If we take the expected value of aggregate surplus as our welfare measure, we therefore see from these two examples that either policy instrument may be preferable, depending on the circumstances.²⁰ (Exercise 11.E.1 asks you to provide a full analysis for a linear-quadratic example.) Note also that the bargaining procedure we have discussed will not result in optimality in *either* case depicted in Figure 11.E.4.²¹ Thus, we have here two cases in which either a quota or a tax performs better than a particular decentralized outcome.²²

20. In Chapter 13, we discuss in greater detail some of the issues that arise in making welfare comparisons in settings with privately held information. There we shall justify the maximization of expected aggregate surplus in this partial equilibrium setting as a requirement of a notion of ex ante Pareto optimality for the two agents. See also the discussion in Section 23.F.

21. Strictly speaking, our previous discussion of bargaining assumed only two possible levels of the externality, while here we have a continuum of levels. This difference is not important. The inefficiency of the bargaining procedure previously studied would hold in this continuous environment as well.

22. We should emphasize that in these two examples other bargaining procedures will perform better than the procedure involving a take-it-or-leave-it offer by the consumer. For example, if a take-or-leave offer is made by the firm, then full optimality results in *both* of these cases because the type of the consumer is known with certainty. The conclusion of our discussion is therefore a qualitative one: With asymmetric information, it is difficult to make very general assertions about the relative performance of centralized versus decentralized approaches.

**Figure 11.E.4**

Two cases in which a quota or tax maximizes aggregate surplus for every realization of θ . (a) Quota $\bar{h} = h^*$ maximizes aggregate surplus for all θ . (b) Tax $t = t^*$ maximizes aggregate surplus for all θ .

In Exercise 11.E.2, you are asked to extend the analysis just given to a case with two firms ($j = 1, 2$) generating an externality, where the two firms are identical except possibly for their realized levels of θ_j . The exercise illustrates the importance of the degree of correlation between the θ_j 's for the relative performance of quotas versus taxes. In comparing a uniform quota policy versus a uniform tax policy ("uniform" here means that the two firms face the same quota or tax rate), the less correlated the shocks across the firms, the better the tax looks. The reason is not difficult to discern. With imperfect correlation, a uniform tax has a benefit that is not achieved with a uniform quota: It allows for the individual levels of externalities generated to be responsive to the realized values of the θ_j 's. Indeed, with a uniform tax, the production of the total amount of externality generated is always efficiently distributed across the two firms.

The presence of multiple generators of an externality also raises the possibility that a market for tradeable emissions permits could be created, as discussed at the end of Section 11.D. This simple addition to the quota policy can potentially eliminate the inefficient distribution of externality generation across different generators that is often a feature of a quota policy. In particular, suppose that instead of simply giving each firm a quota level, we now give them tradeable externality permits entitling them to generate the same number of units of the externality as in the quota. Suppose also that each firm would always fully use its quota if no trade was possible. Then trade must result in *at least* as large a value of aggregate surplus as the simple quota scheme for any realization of the firms' and consumer's types, because we still get the same total level of emissions and we can never get a trade between firms that lowers aggregate profits.²³ Of course, the same bargaining problems that we studied above can prevent a fully efficient distribution of externality generation from arising; but if the firms know each others' values of θ_j , or are numerous enough to act competitively in the market for these rights, then we can expect a distribution of the total externality generation that is efficient across generators. In fact, in the case where the statistical distribution of costs among the firms is known but the particular realizations for individual firms are not known, this type of policy can achieve a fully optimal outcome.

23. Note, however, that the assumption that the externalities generated by the different firms are perfect substitutes to the consumer is crucial to this conclusion. If this is not true, then the reallocation of externality generation can reduce aggregate surplus by lowering the well-being of the agents affected by the externality.

More General Policy Mechanisms

The tax and quota schemes considered above are, as we have seen, completely unresponsive to changes in the marginal costs of the externality to the affected agent (the consumer in this case). It is natural to wonder whether any other sorts of schemes can do better, perhaps by making the level of the externality responsive to the consumer's costs. The problem in doing so is that these benefits and costs are unobservable, and the parties involved may not have incentives to reveal them truthfully if asked. For example, suppose that the government simply asks the consumer and the firm to report their benefits and costs from the externality and then enforces whatever appears to be the optimal outcome based on these reports. In this case, the consumer will have an incentive to exaggerate her costs when asked in order to prevent the firm from being allowed to generate the externality. The question, then, is how to design mechanisms that control these incentives for misreporting and, as a consequence, enable the government to achieve an efficient outcome. This problem is studied in a very general form in Chapter 23; here we confine ourselves to a brief examination of one well-known scheme.

Return to the case in which there are only two possible levels of the externality, 0 and $\bar{h}$. Can we design a scheme that achieves the optimal level of externality generation for every realization of b (the firm's benefit from the externality) and c (the consumer's cost)? We now verify that the answer is "yes."

Imagine the government setting up the following *revelation mechanism*: The firm and the consumer are each asked to report their values of b and c , respectively. Let $\hat{b}$ and $\hat{c}$ denote these announcements. For each possible pair of announcements $(\hat{b}, \hat{c})$, the government sets an allowed level of the externality as well as a tax or subsidy payment for each of the two agents. Suppose, in particular, that the government declares that it will set the allowed externality level h to maximize aggregate surplus given the announcements. That is, $h = \bar{h}$ if and only if $\hat{b} > \hat{c}$. In addition, if externality generation is allowed (i.e., if $h = \bar{h}$), the government will tax the firm an amount equal to $\hat{c}$ and will subsidize the consumer with a payment equal to $\hat{b}$. That is, if the firm wants to generate the externality (which it indicates by reporting a large value of b), it is asked to pay the externality's cost as declared by the consumer; and if the consumer allows the externality (by reporting a low value of c) she receives a payment equal to the externality's benefit as declared by the firm.

In fact, *under this scheme both the firm and the consumer will tell the truth*, so that an optimal level of externality generation will, indeed, result for every possible (b, c) pair. To see this, consider the consumer's optimal announcement when her cost level is c . If the firm announces some $\hat{b} > c$, then the consumer prefers to have the externality-generating activity allowed (she does $\hat{b} - c$ better than if it is prevented). Hence, her optimal announcement satisfies $\hat{c} < \hat{b}$; moreover, because any such announcement will give her the same payoff, she might as well announce the truth, that is, $\hat{c} = c < \hat{b}$. On the other hand, if the firm announces $\hat{b} \leq c$, the consumer prefers to have the externality level set to zero. Hence, she would like announce $\hat{c} \geq \hat{b}$; and again, because any of these announcements will give her the same payoff, she may as well announce the truth, that is, $\hat{c} = c \geq \hat{b}$. Thus, whatever the firm's announcement, truth-telling is an optimal strategy for the consumer. (Formally, telling the truth is a *weakly dominant strategy* for the consumer in the sense studied

in Section 8.B. In fact, it is the consumer's *only* weakly dominant strategy; see Exercise 11.E.3.) A parallel analysis yields the same conclusion for the firm.

Exercise 11.E.4: Show that in the tax-subsidy part of the mechanism above we could add, without affecting the mechanism's truth-telling or optimality properties, an additional payment to each agent that depends in an arbitrary way on the other agent's announcement.

The scheme we have described here is an example of the *Groves–Clarke mechanism* [due to Groves (1973) and Clarke (1971); see also Section 23.C] and was originally proposed as a mechanism for deciding whether to carry out public good projects. Some examples for the public goods context are contained in the exercises at the end of the chapter.

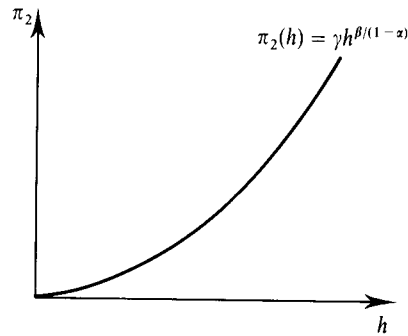
The Groves–Clarke mechanism has two very attractive features: it implements the optimal level of the externality for every (b, c) pair, and it induces truth-telling in a very strong (i.e., dominant strategy) sense. But the mechanism has some unattractive features as well. In particular, it does not result in a balanced budget for the government: The government has a deficit equal to $(b - c)$ whenever $b > c$. We could use the flexibility offered by Exercise 11.E.4 to eliminate this deficit for all possible (b, c) , but then we would necessarily create a budget surplus and therefore a Pareto inefficient outcome for some values of (b, c) (not all units of the numeraire will be left in the hands of the firm or the consumer).

In fact, this problem is unavoidable with this type of mechanism: If we want to preserve the properties that, for every (b, c) , truth-telling is a dominant strategy and the optimal level of externality is implemented, then we generally cannot achieve budget balance for every (b, c) . In Chapter 23 we discuss this issue in greater detail and also consider other mechanisms that can, under certain circumstances, get around the problem. (See also Exercise 11.E.5 for an analysis in which budget balance is required only on average.)

APPENDIX A: NONCONVEXITIES AND THE THEORY OF EXTERNALITIES

Throughout this chapter, we have maintained the assumption that preferences and production sets are convex, leading the derived utility and profit functions we have considered to be concave. With these assumptions, all the decision problems we have studied have been well behaved; they had unique solutions (or, more generally, convex-valued solutions) that varied continuously with the underlying parameters of the problems (e.g., the prices of the L traded commodities or the price of the externality if a market existed for it). Yet, this is not a completely innocent assumption. In this appendix, we present some simple examples designed to illustrate that externalities may themselves generate nonconvexities, and we comment on some of the implications of this fact.

We consider here a bilateral externality situation involving two firms. We suppose that firm 1 may engage in an externality-generating activity that affects firm 2's production. The level of externality generated by firm 1 is denoted by h , and firm j 's profits conditional on the production of externality level h are $\pi_j(h)$ for $j = 1, 2$. It is perfectly natural to assume that $\pi_1(\cdot)$ is concave: The level h could, for example,

**Figure 11.AA.1**

The derived profit function of firm 2 (the externality recipient) in Example 11.AA.1 when $\alpha + \beta > 1$.

be equal to firm 1's output.²⁴ As Examples 11.AA.1 and 11.AA.2 illustrate, however, this may not be true of firm 2's profit function.

Example 11.AA.1: Positive Externalities as a Source of Increasing Returns. Suppose that firm 2 produces an output whose price is 1, using an input whose price, for simplicity, we also take to equal 1. Firm 2's production function is $q = h^\beta z^\alpha$, where $\alpha, \beta \in [0, 1]$. Thus, the externality is a positive one.²⁵ Note that, for fixed h , the problem of firm 2 is concave and perfectly well behaved. Given a level of h , the maximized profits of firm 2 can be calculated to be $\pi_2(h) = \gamma h^{\beta/(1-\alpha)}$, where $\gamma > 0$ is a constant. In Figure 11.AA.1, we represent $\pi_2(h)$ for $\beta > 1 - \alpha$. We see there that firm 2's derived profit function is *not* concave in h ; in fact, it is convex. This reflects the fact that if we think of the externality h as an input to firm 2's production process, then firm 2's overall production function exhibits increasing returns to scale because $\alpha + \beta > 1$. ■

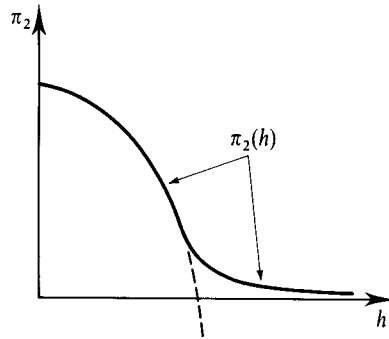
Example 11.AA.2: Negative Externalities as a Source of Nonconvexities. In Example 11.AA.1, the nonconvexity in firm 2's production set, and the resulting failure of concavity in its derived profit function, were caused by a positive externality. In this example the failure of concavity of firm 2's derived profit function is the result of a negative externality.

Suppose, in particular, that $\pi'_2(h) \leq 0$ for all h , with strict inequality for some h , and that firm 2 has the option of shutting down when experiencing externality level h and receiving profits of zero.²⁶ In this case, the function $\pi_2(\cdot)$ can *never* be concave

24. Note also that we may well have $\pi_1(h) < 0$ for some levels $h \geq 0$ because $\pi_1(h)$ is firm 1's maximized profit *conditional* on producing externality level h (and so shutting down is not possible if $h > 0$).

25. More generally, we could think that there is an industry composed of many firms and that the externality is produced and felt by all firms in the industry (e.g., h could be an index, correlated with output, of accumulated know-how in the industry). Externalities were first studied by Marshall (1920) in this context. See also Chipman (1970) and Romer (1986).

26. In the more typical case of a multilateral externality, the ability of affected parties to shut down in this manner often depends on whether the externality is depletable. In the case of a nondepletable externality, such as air pollution, affected firms can always shut down and receive zero profits. In contrast, in the case of a depletable externality (such as garbage), where $\pi_j(h)$ reflects firm j 's profits when it individually absorbs h units of the externality, the absorption of the externality may itself require the use of some inputs (e.g., land to absorb garbage). Indeed, were this not the case for a depletable externality, the externality could always be absorbed in a manner that creates no social costs by allocating all of the externality to a firm that shuts down.

**Figure 11.AA.2**

If the recipient of a negative externality can shut down and earn zero profits for any level of the externality, then its derived profit function $\pi_2(h)$ cannot be concave over $h \in [0, \infty]$.

over all $h \in [0, \infty)$, a point originally noted by Starrett (1972). The reason can be seen in Figure 11.AA.2: If $\pi_2(\cdot)$ were a strictly decreasing concave function, then it would have to become negative at some level of h (see the dashed curve), but $\pi_2(\cdot)$ must be nonnegative if firm 2 can always choose to shut down. ■

The failure of $\pi_2(\cdot)$ to be concave can create problems for both centralized and decentralized solutions to the externality problem. For example, if property rights over the externality are defined and a market for the externality is introduced in either Example 11.AA.1 or Example 11.AA.2, a competitive equilibrium may fail to exist (even assuming that the two agents act as price takers). Firm 2's objective function will not be concave, and so its optimal demand may fail to be well defined and continuous (recall our discussion in Section 10.C of the equilibrium existence problems caused by nonconvexities in firms' cost functions).

In contrast, taxes and quotas can, in principle, still implement the optimal outcome despite the failure of firm 2's profit function to be concave because their use depends only on the profit function of the externality generator (here, firm 1) being well behaved. In practice, however, nonconvexities in firm 2's profit function may create problems for these centralized solutions as well. Example 11.AA.3 illustrates this point.

Example 11.AA.3: Externalities as a Source of Multiple Local Social Optima. It is, in principle, true that if the decision problem of the generator of an externality is concave, then the optimum can be sustained by means of quotas or taxes. But if $\pi_2(\cdot)$ is not concave, then the aggregate surplus function $\pi_1(h) + \pi_2(h)$ may not be concave and, as a result, the first-order conditions for aggregate surplus maximization may suffice only for determining local optima. In fact, as emphasized by Baumol and Oates (1988), the nonconvexities created by externalities may easily generate situations with multiple local social optima, so that identifying a global optimum may be a formidable task.

Suppose, for example, that the profit functions of the two firms are

$$\pi_1(h) = \begin{cases} h & \text{for } h \leq 1 \\ 1 & \text{for } h > 1 \end{cases}$$

and

$$\pi_2(h) = \begin{cases} 2(1-h)^2 & \text{for } h \leq 1 \\ 0 & \text{for } h > 1. \end{cases}$$

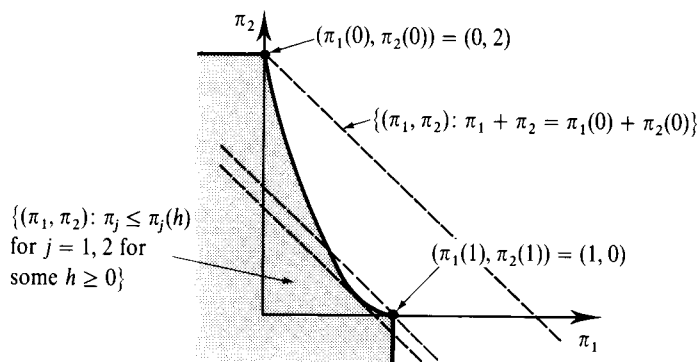


Figure 11.AA.3

The set of possible profit pairs (π_1, π_2) in Example 11.AA.3 exhibits multiple local maxima of aggregate surplus $\pi_1(h) + \pi_2(h)$.

The function $\pi_2(\cdot)$ is not concave, something that the two previous examples have shown us can easily happen with externalities. The profit levels for the two firms that are attainable for different levels of h are depicted in Figure 11.AA.3 by the shaded set $\{(\pi_1, \pi_2): \pi_j \leq \pi_j(h) \text{ for } j = 1, 2 \text{ for some } h > 0\}$ (note that this definition allows for free disposal of profits). The social optimum has $h = 0$ (joint profits are then equal to 2), in which case firm 2 is able to operate in an environment free from the externality. This can be implemented by setting a tax rate on firm 1 of $t > 1$ per unit of the externality. But note that the outcome $h = 1$ (implemented by setting a tax rate on firm 1 of $t = 0$) is a local social optimum: As we decrease h , it is not until $h < \frac{1}{2}$ that we get an aggregate surplus level higher than that at $h = 1$. Hence, this latter outcome satisfies both the first-order and second-order conditions for the maximization of aggregate surplus (e.g., at this point, the marginal benefits of the externality exactly equal its marginal costs), and it will be easy for a social planner to be misled into thinking that she is at a welfare maximum. ■

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EXERCISES

11.B.1^B (M. Weitzman) On Farmer Jones' farm, only honey is produced. There are two ways to make honey: with and without bees. A bucket full of artificial honey, absolutely indistinguishable from the real thing, is made out of 1 gallon of maple syrup with one unit of labor. If the same honey is made the old-fashioned way (with bees), k total units of labor are required (including bee-keeping) and b bees are required per bucket. Either way, Farmer Jones has the capacity to produce up to H buckets of honey on his farm.

The neighboring farm, belonging to Smith, produces apples. If bees are present, less labor is needed because bees pollinate the blossoms instead of workers doing it. For this reason, c bees replace one worker in the task of pollinating. Up to A bushels of apples can be grown on Smith's farm.

Suppose that the market wage rate is w , bees cost p_b per bee, and maple syrup costs p_m per gallon. If each farmer produces her maximal output at the cheapest cost to her (assume the output prices they face make maximal production efficient), is the resulting outcome efficient? How does the answer depend on k , b , c , w , p_b , and p_m ? Give an intuitive explanation of your result. Up to how much would Smith be willing to bribe Jones to produce honey with bees? What would happen to efficiency if both farms belonged to the same owner? How could the government achieve efficient production through taxes?

11.B.2^C Consider the two-consumer externality problem studied in Section 11.B, but now assume that consumer 2's derived utility function over the externality level h and her wealth available for commodity purchases w_2 takes the form $\phi_2(h, w_2)$. Assume that $\phi_2(h, w_2)$ is a twice-differentiable, strictly quasiconcave function with $\partial\phi_2(h, w_2)/\partial w_2 > 0$ and, for simplicity, that we have a positive externality so that $\partial\phi_2(h, w_2)/\partial h > 0$.

(a) Set up the Pareto optimality problem as one of choosing h and a wealth transfer T to maximize consumer 1's welfare subject to giving consumer 2 a utility level of at least $\bar{u}_2$. Derive the (necessary and sufficient) first-order condition characterizing the optimal levels of h and T , say h° and T° .

(b) Imagine that consumer 1 could purchase h on an externality market. Let p_h be the price per unit, and let $h(p_h, w_2)$ be consumer 2's demand function for h . Express the wealth effect $\partial h(p_h, w_2)/\partial w_2$ in terms of first-order and second-order partial derivatives of consumer 2's utility function.

(c) Derive the comparative statics change in the Pareto optimal level of the externality h° (for a given $\bar{u}_2$) with respect to a differential increase $dw_2 > 0$ in consumer 2's wealth. Show that if consumer 2's demand for the externality, derived in (b), is normal at price $\bar{p}_h = [\partial\phi_2(h^\circ, w_2 - T^\circ)/\partial h] / [\partial\phi_2(h^\circ, w_2 - T^\circ)/\partial w_2]$ and wealth level $\bar{w}_2 = w_2 - T^\circ$ [i.e., if $\partial h(\bar{p}_h, \bar{w}_2)/\partial w_2 > 0$], then a marginal increase in consumer 2's wealth w_2 causes the Pareto optimal level of the externality h° to increase. (Similarly, in the case of a negative externality, if consumer 2's demand for reductions in the externality is a normal good, then when consumer 2 becomes wealthier, the Pareto optimal level of the externality declines.)

11.B.3^B Consider the optimal Pigouvian tax identified in Section 11.B for the two-consumer externality problem studied there. What happens if, given this tax, the two consumers are able to bargain with each other? Will the efficient level of the externality still result? What about with the optimal quota?

11.B.4^B Consider again the two-consumer externality problem studied in Section 11.B. Suppose that consumer 2 can take some action, say $e \in \mathbb{R}$, that affects the degree to which she is affected by the externality, so that we now write her derived utility function as $\phi_2(h, e) + w_2$. To fix ideas, let h be a negative externality, and suppose that $\partial^2\phi_2(h, e)/\partial h\partial e > 0$, so that increases in e reduce the negative effect of the externality on the margin. Suppose that both h and e can in principle be taxed or subsidized. Should e be taxed or subsidized in the optimal tax scheme? Why or why not?

11.B.5^B Suppose that at fixed input prices of $\bar{w}$ a firm produces output with the differentiable and strictly convex cost function $c(q, h)$, where $q \geq 0$ is its output level (whose price is $p > 0$) and h is the level of a negative externality generated by the firm. The externality affects a single consumer, whose derived utility function takes the form $\phi(h) + w$. The actions of the firm and consumer do not affect any market prices.

(a) Derive the first-order condition for the firm's choice of q and h .

(b) Derive the first-order conditions characterizing the Pareto optimal levels of q and h .

(c) Suppose that the government taxes the firm's output level. Show that this cannot restore efficiency. Show that a direct tax on the externality *can* restore efficiency.

(d) Show, however, that in the limiting case where h is necessarily produced in fixed proportions with q , so that $h(q) = \alpha q$ for some $\alpha > 0$, a tax on the firm's output *can* restore efficiency. What is the efficiency-restoring tax?

11.C.1^A Consider the model discussed in Section 11.C, in which I consumers privately purchase a public good. Identify per-unit subsidies $s_1, \dots, s_I$, such that when each consumer i faces subsidy rate s_i , the total level of the public good provided is optimal.

11.C.2^A Consider the model discussed in Section 11.C, in which I consumers privately purchase a public good. Show that a per-unit subsidy on the firm's output (paid to the firm) can also restore efficiency.

11.C.3^C Reconsider the Ramsey tax problem from Exercise 10.E.3, but now suppose that the government can also provide a public good x_0 that can be produced from good 1 at cost $c(x_0)$. However, the government must still balance its budget (including any expenditures on the public good). Consumer i 's utility function now takes the form $x_{1i} + \sum_{\ell=2}^L \phi_{\ell i}(x_{\ell i}, x_0)$. Derive and interpret the conditions characterizing the optimal commodity taxes and the optimal level of the public good. How do the two problems of Ramsey taxation and provision of the public good interact?

11.D.1^B (M. Weitzman) First-year graduate students are a hard-working group. Consider a typical class of I students. Suppose that each student i puts in h_i hours of work on her classes. This effort involves a disutility of $h_i^2/2$. Her benefits depend on how she performs relative to her peers and take the form $\phi(h_i/\bar{h})$ for all i , where $\bar{h} = (1/I)\sum_i h_i$ is the average number of hours put in by all students in the class and $\phi(\cdot)$ is a differentiable concave function, with

$\phi'(\cdot) > 0$ and $\lim_{h \rightarrow 0} \phi'(h) = \infty$. Characterize the symmetric (Nash) equilibrium. Compare it with the Pareto optimal symmetric outcome. Interpret.

11.D.2^B Consider a setting with I consumers. Each consumer i chooses an action $h_i \in \mathbb{R}_+$. Consumer i 's derived utility function over her choice of h and the choices of other consumers takes the form $\phi_i(h_i, \sum_{i \neq i} h_i) + w_i$, where $\phi_i(\cdot)$ is strictly concave. Characterize the optimal levels of $h_1, \dots, h_I$. Compare these with the equilibrium levels. What tax/subsidy scheme induces the optimal outcome?

11.D.3^B Consider an industry composed of $J > 1$ identical firms that act as price takers. The price of their output is p , and the prices of their inputs are unaffected by their actions. Suppose that partial equilibrium analysis is valid and that the aggregate demand for their product is given by the function $x(p)$. The industry is characterized by "learning by doing," in that each firm's total cost of producing a given level of output is declining in the level of total industry output; that is, each firm j has a twice-differentiable cost function of the form $c(q_j, Q)$ for $Q = \sum_j q_j$, where $c(\cdot)$ is strictly increasing in its first argument and strictly decreasing in its second. Letting subscripts denote partial derivatives, assume that $c_q + Jc_Q > 0$ and $(1/n)c_{qq} + 2c_{qQ} + nc_{QQ} > 0$ for $n = 1$ and J . Compare the equilibrium and optimal industry output levels. Interpret. What tax or subsidy restores efficiency?

11.D.4^B Reconsider the nondepletable externality example discussed in Section 11.D, but now assume that the externalities produced by the J firms are not homogeneous. In particular, suppose that if $h_1, \dots, h_J$ are the firms' externality levels, then consumer i 's derived utility is given by $\phi_i(h_1, \dots, h_J) + w_i$ for each $i = 1, \dots, I$. Compare the equilibrium and efficient levels of $h_1, \dots, h_J$. What tax/subsidy scheme can restore efficiency? Under what condition should each firm face the same tax/subsidy rate?

11.D.5^B (*The problem of the commons*) Lake Ec can be freely accessed by fishermen. The cost of sending a boat out on the lake is $r > 0$. When b boats are sent out onto the lake, $f(b)$ fish are caught in total [so each boat catches $f(b)/b$ fish], where $f'(b) > 0$ and $f''(b) < 0$ at all $b \geq 0$. The price of fish is $p > 0$, which is unaffected by the level of the catch from Lake Ec.

- Characterize the equilibrium number of boats that are sent out on the lake.
- Characterize the optimal number of boats that should be sent out on the lake. Compare this with your answer to (a).
- What per-boat fishing tax would restore efficiency?
- Suppose that the lake is instead owned by a single individual who can choose how many boats to send out. What level would this owner choose?

11.D.6^B Suppose that there is a piece of land that is affected adversely by an externality produced by a single firm. The firm's derived profit function for the externality is $\pi(h) = \alpha + \beta h - \mu h^2$, where h is the level of the externality and $(\alpha, \beta, \mu) \gg 0$. There are I consumers who farm the land, each owning a fraction $1/I$ of it. The total yield of the land is $\phi(h) = \gamma - \eta h$, where $(\gamma, \eta) \gg 0$. Each of the I consumers then has a derived utility function of $\phi(h)/I + w$.

Bargaining among the consumers and the firm works as follows: Each consumer simultaneously decides whether to be in or out of a bargaining coalition. After this, the bargaining coalition makes the firm a take-it-or-leave-it offer specifying a level of h and a transfer. The firm then accepts or rejects this offer. In the absence of any agreement, the firm can generate any level of the externality it wishes.

- Let θ denote the fraction of the I consumers who join the bargaining coalition. Characterize the subgame perfect Nash equilibrium level of θ (for simplicity, treat θ as a continuous variable). Show that when $I = 1$ the optimal level of the externality results, but that when $I > 1$ we have $\theta < 1$ in equilibrium and too much of the externality is generated.

(b) Show that as I increases, the equilibrium level of θ declines. Also show that $\lim_{I \rightarrow \infty} \theta = 0$.

11.D.7^C Individuals can build their houses in one of two neighborhoods, A or B. It costs c_A to build a house in neighborhood A and $c_B < c_A$ to build in neighborhood B. Individuals care about the prestige of the people living in their neighborhood. Individuals have varying levels of prestige, denoted by the parameter θ . Prestige varies between 0 and 1 and is uniformly distributed across the population. The prestige of neighborhood k ($k = A, B$) is a function of the average value of θ in that neighborhood, denoted by $\bar{\theta}_k$. If individual i has prestige parameter θ and builds her house in neighborhood k , her derived utility net of building costs is $(1 + \theta)(1 + \bar{\theta}_k) - c_k$. Thus, individuals with more prestige value a prestigious neighborhood more. Assume that c_A and c_B are less than 1 and that $c_A - c_B \in (\frac{1}{2}, 1)$.

(a) Show that in any building-choice equilibrium (technically, the Nash equilibrium of the simultaneous-move game in which individuals simultaneously choose where to build their house) both neighborhoods must be occupied.

(b) Show that in any equilibrium in which the prestige levels of the two neighborhoods differ, every resident of neighborhood A must have at least as high a prestige level as every resident of neighborhood B; that is, there is a cutoff level of θ , say $\hat{\theta}$, such that all types $\theta \geq \hat{\theta}$ build in neighborhood A and all $\theta < \hat{\theta}$ build in neighborhood B. Characterize this cutoff level.

(c) Show that in any equilibrium of the type identified in (b), a Pareto improvement can be achieved by altering the cutoff value of θ slightly.

11.E.1^B Consider the setting studied in Section 11.E, and suppose that $\partial\pi(h, \theta)/\partial h = \beta - bh + \theta$ and $\partial\phi(h, \eta)/\partial h = \gamma - ch + \eta$, where θ and η are random variables with $E[\theta] = E[\eta] = E[\theta\eta] = 0$, $(\beta, b, c) \gg 0$, and $\gamma < 0$. Denote $E[\theta^2] = \sigma_\theta^2$ and $E[\eta^2] = \sigma_\eta^2$.

(a) Identify the best quota $\hat{h}^*$ for a planner who wants to maximize the expected value of aggregate surplus. (Assume the firm must produce an amount exactly equal to the quota.)

(b) Identify the best tax t^* for this same planner.

(c) Compare the two instruments: Which is better and when?

11.E.2^C Extend the model in Exercise 11.E.1 to the case of two producers. Now let $\partial\pi_i(h_i, \theta_i)/\partial h = \beta - bh_i + \theta_i$ for $i = 1, 2$. Let $\sigma_{12} = E[\theta_1\theta_2]$. Calculate and compare the optimal quotas and taxes. How does the choice depend on σ_{12} ?

11.E.3^B Show that truth-telling is the consumer's only weakly dominant strategy in the (Groves–Clarke) revelation mechanism studied in Section 11.E.

11.E.4^A In text.

11.E.5^B Suppose that the government is considering building a public project. The cost is K . There are I individuals indexed by i . Individual i 's privately known benefit from the project is b_i . The government's objective is to maximize the expected value of aggregate surplus. Derive the extension of the Groves–Clarke mechanism discussed in Section 11.E for this case. Can you construct your scheme so that the government balances its budget on average (over all realizations of the b_i 's)?

11.E.6^B Extend Exercise 11.E.5 to the case in which there are N possible projects, $n = 1, \dots, N$, with individual i deriving a (privately known) benefit of $b_i(n)$ from project n .

11.E.7^B Suppose that in the model of Section 11.E the consumer's type η takes only one possible value, $\bar{\eta}$. We have seen in the text that in this case neither a quota nor a tax will maximize aggregate surplus for all realizations of θ when the derived utility function $\phi(h, \bar{\eta})$ for the consumer has $\partial\phi(h, \bar{\eta})/\partial h \in (0, -\infty)$. Show, however, that a variable tax per unit in which the total tax collected from the firm is $\phi(h, \bar{\eta})$ when the level of the externality is h will maximize aggregate surplus for all values of θ for *any* derived utility function $\phi(h, \bar{\eta})$.

Market Power

12.A Introduction

In the competitive model, all consumers and producers are assumed to act as price takers, in effect behaving as if the demand or supply functions that they face are infinitely elastic at going market prices. However, this assumption may not be a good one when there are only a few agents on one side of a market, for these agents will often possess *market power*—the ability to alter profitably prices away from competitive levels.

The simplest example of market power arises when there is only a single seller, a *monopolist*, of some good. If this good's market demand is a continuous decreasing function of price, then the monopolist, recognizing that a small increase in its price above the competitive level leads to only a small reduction in its sales, will find it worthwhile to raise its price above the competitive level.

Similar effects can occur when there is more than one agent, but still not many, on one side of a market. Most often, these agents with market power are firms, whose fewness arises from nonconvexities in production technologies (recall the discussion of entry in Section 10.F).

In this chapter, we study the functioning of markets in which market power is present. We begin, in Section 12.B, by considering the case in which there is a monopolist seller of some good. We review the theory of monopoly pricing and identify the welfare loss that it creates.

The remaining sections focus on situations of *oligopoly*, in which a number of firms compete in a market. In Sections 12.C and 12.D, we discuss several models of oligopolistic pricing. Each incorporates different assumptions about the underlying structure of the market and behavior of firms. The discussion highlights the implications of these differing assumptions for market outcomes. In Section 12.C, we focus on static models of oligopolistic pricing, where competition is viewed as a one-shot, simultaneous event. In contrast, in Section 12.D, we study how repeated interaction among firms may affect pricing in oligopolistic markets. This discussion constitutes an application of the theory of repeated games, a subject that we discuss in greater generality in Appendix A.

The analysis in Sections 12.B to 12.D treats the number of firms in the market as

exogeneously given. In reality, however, the number of active firms in a market is likely to be affected by factors such as the size of market demand and the nature of competition within the market. Sections 12.E and 12.F consider issues that arise when the number of active firms in a market is determined endogenously.

Section 12.E specifies a simple model of entry into an oligopolistic market and studies the determinants of the number of active firms. It offers an analysis that parallels that considered in Section 10.F for competitive markets.

Section 12.F returns to a theme raised in Chapter 10. We illustrate how the competitive (price-taking) model can be viewed as a limiting case of oligopoly in which the size of the market, and hence the number of firms that can profitably operate in it, grows large. In the model we study, an active firm's market power diminishes as the market size expands; in the limit, the equilibrium market price comes to approximate the competitive level.

In Section 12.G, we briefly consider how firms in oligopolistic markets can make strategic precommitments to affect the conditions of future competition in a manner favorable to themselves. This issue nicely illustrates the importance of credible commitments in strategic settings, an issue we studied extensively in Chapter 9. In Appendix B, we consider in greater detail a particularly striking example of strategic precommitment to affect future market conditions, the case of entry deterrence through capacity choice.

If you have not done so already, you should review the game theory chapters in Part II before studying Sections 12.C to 12.G (in particular, review all of Chapter 7, Sections 8.A to 8.D, and Sections 9.A and 9.B).

An excellent source for further study of the topics covered in this chapter is Tirole (1988).¹

12.B Monopoly Pricing

In this section, we study the pricing behavior of a profit-maximizing *monopolist*, a firm that is the only producer of a good. The demand for this good at price p is given by the function $x(p)$, which we take to be continuous and strictly decreasing at all p such that $x(p) > 0$.² For convenience, we also assume that there exists a price $\bar{p} < \infty$ such that $x(p) = 0$ for all $p \geq \bar{p}$.³ Throughout, we suppose that the monopolist knows the demand function for its product and can produce output level q at a cost of $c(q)$.

The monopolist's decision problem consists of choosing its price p so as to maximize its profits (in terms of the numeraire), or formally, of solving

$$\max_p px(p) - c(x(p)). \quad (12.B.1)$$

1. See also the survey by Shapiro (1989) for the topics covered in Sections 12.C, 12.D, and 12.G.

2. Throughout this chapter we take a partial equilibrium approach; see Chapter 10 for a discussion of this approach.

3. This assumption helps to insure that an optimal solution to the monopolist's problem exists. (See Exercise 12.B.2 for an example in which the failure of this condition leads to nonexistence.)

An equivalent formulation in terms of quantity choices can be derived by thinking instead of the monopolist as deciding on the level of output that it desires to sell, $q \geq 0$, letting the price at which it can sell this output be given by the *inverse demand function* $p(\cdot) = x^{-1}(\cdot)$.⁴ Using this inverse demand function, the monopolist's problem can then be stated as

$$\text{Max}_{q \geq 0} \quad p(q)q - c(q). \quad (12.B.2)$$

We shall focus our analysis on this quantity formulation of the monopolist's problem [identical conclusions could equally well be developed from problem (12.B.1)]. We assume throughout that $p(\cdot)$ and $c(\cdot)$ are continuous and twice differentiable at all $q \geq 0$, that $p(0) > c'(0)$, and that there exists a unique output level $q^\circ \in (0, \infty)$ such that $p(q^\circ) = c'(q^\circ)$. Thus, q° is the unique socially optimal (competitive) output level in this market (see Chapter 10).

Under these assumptions, a solution to problem (12.B.2) can be shown to exist.⁵ Given the differentiability assumed, the monopolist's optimal quantity, which we denote by q^m , must satisfy the first-order condition⁶

$$p'(q^m)q^m + p(q^m) \leq c'(q^m), \quad \text{with equality if } q^m > 0. \quad (12.B.3)$$

The left-hand side of (12.B.3) is the *marginal revenue* from a differential increase in q at q^m , which is equal to the derivative of revenue $d[p(q)q]/dq$, while the right-hand side is the corresponding marginal cost at q^m . Since $p(0) > c'(0)$, condition (12.B.3) can be satisfied only at $q^m > 0$. Hence, under our assumptions, *marginal revenue must equal marginal cost* at the monopolist's optimal output level:

$$p'(q^m)q^m + p(q^m) = c'(q^m). \quad (12.B.4)$$

For the typical case in which $p'(q) < 0$ at all $q \geq 0$, condition (12.B.4) implies that we must have $p(q^m) > c'(q^m)$, and so *the price under monopoly exceeds marginal cost*. Correspondingly, the monopolist's optimal output q^m must be below the socially optimal (competitive) output level q° . The cause of this quantity distortion is the monopolist's recognition that a reduction in the quantity it sells allows it to increase the price charged on its remaining sales, an increase whose effect on profits is captured by the term $p'(q^m)q^m$ in condition (12.B.4).

The welfare loss from this quantity distortion, known as the *deadweight loss of monopoly*, can be measured using the change in Marshallian aggregate surplus

4. More precisely, to take account of the fact that $x(p) = 0$ for more than one value of p , we take $p(q) = \text{Min} \{p : x(p) = q\}$ at all $q \geq 0$. Thus, $p(0) = \bar{p}$, the lowest price at which $x(p) = 0$.

5. In particular, it follows from condition (12.B.3) and from the facts that $p'(q) \leq 0$ for all $q \geq 0$ and $p(q) < c'(q)$ for all $q > q^\circ$, that the monopolist's optimal choice must lie in the compact set $[0, q^\circ]$. Because the objective function in problem (12.B.2) is continuous, a solution must therefore exist (see Section M.F of the Mathematical Appendix).

6. Satisfaction of first-order condition (12.B.3) is sufficient for q^m to be an optimal choice if the objective function of problem (12.B.2) is concave on $[0, q^\circ]$. Note, however, that concavity of this objective function depends not only on the technology of the firm, as in the competitive model, but also on the shape of the inverse demand function. In particular, even with a convex cost function, the monopolist's profit function can violate this concavity condition if demand is a convex function of price.

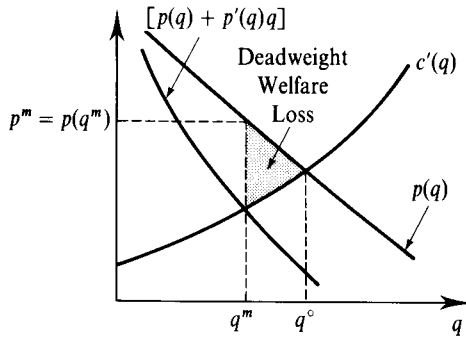


Figure 12.B.1 (left)
The monopoly solution and welfare loss when $p'(\cdot) < 0$.

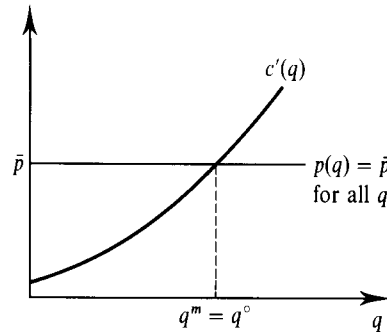


Figure 12.B.2 (right)
The monopoly solution when $p'(q) = 0$ for all q .

(see Section 10.E),

$$\int_{q^m}^{q^o} [p(s) - c'(s)] ds > 0,$$

where q^o is the socially optimal (competitive) output level.

Figure 12.B.1 illustrates the monopoly outcome in this case. The monopolist's quantity q^m is determined by the intersection of the graphs of marginal revenue $p'(q)q + p(q)$ and marginal cost $c'(q)$. The monopoly price $p(q^m)$ can then be determined from the inverse demand curve. The deadweight welfare loss is equal to the area of the shaded region.

Note from condition (12.B.4) that the monopoly quantity distortion is absent in the special case in which $p'(q) = 0$ for all q . In this case, where $p(q)$ equals some constant $\bar{p}$ at all $q > 0$, the monopolist sells the same quantity as a price-taking competitive firm because it perceives that any increase in its price above the competitive price $\bar{p}$ causes it to lose all its sales.⁷ Figure 12.B.2 depicts this special case.

Example 12.B.1: Monopoly Pricing with a Linear Inverse Demand Function and Constant Returns to Scale. Suppose that the inverse demand function in a monopolized market is $p(q) = a - bq$ and that the monopolist's cost function is $c(q) = cq$, where $a > c \geq 0$ [so that $p(0) > c'(0)$] and $b > 0$. In this case, the objective function of the monopolist's problem (12.B.2) is concave, and so condition (12.B.4) is both necessary and sufficient for a solution to the monopolist's problem. From condition (12.B.4), we can calculate the monopolist's optimal quantity and price to be $q^m = (a - c)/2b$ and $p^m = (a + c)/2$. In contrast, the socially optimal (competitive) output level and price are $q^o = (a - c)/b$ and $p^o = p(q^o) = c$. ■

Although we do not discuss these issues here, we point out that the behavioral distortions arising under monopoly are not limited to pricing decisions. (Exercises 12.B.9 and 12.B.10 ask you to investigate two examples.)

The monopoly quantity distortion is fundamentally linked to the fact that if the monopolist wants to increase the quantity it sells, it must lower its price on *all* its existing sales. In fact,

7. This inverse demand function arises, for example, when each consumer i has quasilinear preferences of the form $u_i(q_i) + m_i$ with $u_i(q_i) = \bar{p}q_i$, where q_i is consumer i 's consumption of the good under study and m_i is his consumption of the numeraire commodity. [Strictly speaking, with these preferences we now have a multivalued demand correspondence rather than a demand function, but $p(\cdot)$ is nevertheless a function as before.]

if the monopolist were able to *perfectly discriminate* among its customers in the sense that it could make a distinct offer to each consumer, knowing the consumer's preferences for its product, then the monopoly quantity distortion would disappear.

To see this formally, let each consumer i have a quasilinear utility function of the form $u_i(q_i) + m_i$ over the amount q_i of the monopolist's good that he consumes and the amount m_i that he consumes of the numeraire good, and normalize $u_i(0) = 0$. Suppose that the monopolist makes a take-it-or-leave-it offer to each consumer i of the form (q_i, T_i) , where q_i is the quantity offered to consumer i and T_i is the total payment that the consumer must make in return. Given offer (q_i, T_i) , consumer i will accept the monopolist's offer if and only if $u_i(q_i) - T_i \geq 0$. As a result, the monopolist can extract a payment of exactly $u_i(q_i)$ from consumer i in return for q_i units of its product, leaving the consumer with a surplus of exactly zero from consumption of the good. Given this fact, the monopolist will choose the quantities it sells to the I consumers $(q_1, \dots, q_I)$ to solve

$$\text{Max}_{(q_1, \dots, q_I) \geq 0} \sum_{i=1}^I u_i(q_i) - c(\sum_i q_i). \quad (12.B.5)$$

Note, however, that any solution to problem (12.B.5) maximizes the aggregate surplus in the market, and so the monopolist will sell each consumer exactly the socially optimal (competitive) quantity. Of course, the distributional properties of this outcome would not be terribly attractive in the absence of wealth redistribution: The monopolist would get all the aggregate surplus generated by its product, and each consumer i would receive a surplus of zero (i.e., each consumer i 's welfare would be exactly equal to the level he would achieve if he consumed none of the monopolist's product). But in principle, these distributional problems can be corrected through lump-sum redistribution of the numeraire.

Thus, the welfare loss from monopoly pricing can be seen as arising from constraints that prevent the monopolist from charging fully discriminatory prices. In practice, however, these constraints can be significant. They may include the costs of assessing separate charges for different consumers, the monopolist's lack of information about consumer preferences, and the possibility of consumer resale. Exercise 12.B.5 explores some of these factors. It provides conditions under which the best the monopolist can do is to name a single per-unit price, as we assumed at the beginning of this section.

12.C Static Models of Oligopoly

We now turn to cases in which more than one, but still not many, firms compete in a market. These are known as situations of *oligopoly*. Competition among firms in an oligopolistic market is inherently a setting of strategic interaction. For this reason, the appropriate tool for its analysis is game theory. Because this discussion constitutes our first application of the theory of games, we focus on relatively simple *static* models of oligopoly, in which there is only one period of competitive interaction and firms take their actions simultaneously.

We begin by studying a model of simultaneous price choices by firms with constant returns to scale technologies, known as the *Bertrand model*. This model displays a striking feature: With just two firms in a market, we obtain a perfectly competitive outcome. Motivated by this finding, we then consider three alterations of this model that weaken its strong and often implausible conclusion: a change in the firm's strategy from choosing its price to choosing its quantity of output

(the *Cournot model*); the introduction of capacity constraints (or, more generally, decreasing returns to scale); and the presence of product differentiation.⁸

One lesson of this analysis is that a critical part of game-theoretic modeling goes into choosing the strategies and payoff functions of the players. In the context of oligopolistic markets, this choice requires that considerable thought be given both to the demand and technological features of the market and to the underlying processes of competition.

Unless otherwise noted, we restrict our attention to pure strategy equilibria of the models we study.

The Bertrand Model of Price Competition

We begin by considering the model of oligopolistic competition proposed by Bertrand (1883). There are two profit-maximizing firms, firms 1 and 2 (a *duopoly*), in a market whose demand function is given by $x(p)$. As in Section 10.B, we assume that $x(\cdot)$ is continuous and strictly decreasing at all p such that $x(p) > 0$ and that there exists a $\bar{p} < \infty$ such that $x(p) = 0$ for all $p \geq \bar{p}$. The two firms have constant returns to scale technologies with the same cost, $c > 0$, per unit produced. We assume that $x(c) \in (0, \infty)$, which implies that the socially optimal (competitive) output level in this market is strictly positive and finite (see Chapter 10).

Competition takes place as follows: The two firms simultaneously name their prices p_1 and p_2 . Sales for firm j are then given by

$$x_j(p_j, p_k) = \begin{cases} x(p_j) & \text{if } p_j < p_k \\ \frac{1}{2}x(p_j) & \text{if } p_j = p_k \\ 0 & \text{if } p_j > p_k. \end{cases}$$

The firms produce to order and so they incur production costs only for an output level equal to their actual sales. Given prices p_j and p_k , firm j 's profits are therefore equal to $(p_j - c)x_j(p_j, p_k)$.

The Bertrand model constitutes a well-defined simultaneous-move game to which we can apply the concepts developed in Chapter 8. In fact, the Nash equilibrium outcome of this model, presented in Proposition 12.C.1, is relatively simple to discern.

Proposition 12.C.1: There is a unique Nash equilibrium (p_1^*, p_2^*) in the Bertrand duopoly model. In this equilibrium, both firms set their prices equal to cost: $p_1^* = p_2^* = c$.

Proof: To begin, note that both firms setting their prices equal to c is indeed a Nash equilibrium. At these prices, both firms earn zero profits. Neither firm can gain by raising its price because it will then make no sales (thereby still earning zero); and by lowering its price below c a firm increases its sales but incurs losses. What remains is to show that there can be no other Nash equilibrium.⁹ Suppose, first, that the lower of the two prices named is less than c . In this case, the firm naming this price

8. Section 12.D studies a fourth variation that involves repeated interaction among firms.

9. Recall that we restrict attention to pure strategy equilibria here. See Exercise 12.C.2 for a consideration of mixed strategy equilibria. There you are asked to show that under the conditions assumed here, Proposition 12.C.1 continues to hold: $p_1^* = p_2^* = c$ is the unique Nash equilibrium, pure or mixed, of the Bertrand model.